GJTD-LR: A Trainable Grouped Joint Tensor Dictionary With Low-Rank Prior for Single Hyperspectral Image Super-Resolution

Cong Liu, Zhihao Fan, and Guixu Zhang

***Abstract*— Reconstructing a high-resolution hyperspectral image (HR-HSI) using a single low-resolution hyperspectral image (LR-HSI) is a significant technique for increasing the spatial resolution of HSIs and overcoming the physical limitation of the HSI sensor. Most single HSI super-resolution methods have achieved great success recently. However, owning to the difficulty of acquiring an HSI, the available training samples are relatively few, which will inevitably lead to relatively low performance. To address this issue, in this article, we propose a novel single HSI super-resolution method by combining a trainable grouped joint tensor dictionary and a low-rank prior (GJTD-LR). First, we design a trainable grouped joint tensor dictionary, which can build an accurate mapping relationship between training HR-HSIs and their corresponding LR-HSIs with relatively few training samples. To be specific, the training HR-HSI and LR-HSI pairs are decomposed into a joint tensor dictionary and a set of sparse coefficients using tensor–tensor product to fully preserve the spectral correlation. In addition, we apply a grouped strategy to divide the training images into several groups and learn a compact joint dictionary for each group. Second, a tensor low-rank model is forced into the reconstruction model to further capture the spatial correlation. Finally, GJTD-LR is optimized using alternating direction method of multipliers (ADMM), soft threshold algorithm, singular value decomposition, and Fourier domain transform. The experimental results on both remote sensed HSIs and indoor HSIs show the superiority of GJTD-LR to some other traditional and advanced single HSI super-resolution methods.**

***Index Terms*— Group joint tensor dictionary, hyperspectral image (HSI) super-resolution, low-rank prior, tensor–tensor product.**

I. Introduction

HYPERSPECTRAL imaging technique is an advanced technique that obtains a set of images with hundreds of narrow spectral bands simultaneously for the same scene. The hyperspectral image (HSI) with hundreds of spectral bands provides rich spectral information, which is helpful to identify objects and materials accurately. Therefore, it plays an important role in many applications such as remote sensing, disease diagnosis and various computer vision tasks [1], [2]. However, as we all know, owing to the hardware limitations of the hyperspectral imaging sensor, it is very difficult to get an HSI containing both high spectral resolution and high spatial resolution simultaneously. In other words, there is a tradeoff between the spectral and the spatial resolutions in acquiring an HSI. As a result, the researchers usually maintain the high spectral resolution while sacrificing the spatial resolution to get a low spatial resolution HSI. Thus, it is necessary to search some effective and economical ways to get the high spatial resolution HSI so as to better and accurately describe the objects and materials. HSI super-resolution is such a way that aims to reconstruct a high spatial resolution HSI from one or more corresponding degraded low spatial resolution HSIs. Recently, the research of the HSI super-resolution technique has become a hotspot and a number of studies commit to the study of this field. According to whether the auxiliary images are used, the existing HSI super-resolution methods can be roughly divided into two categories, fusion-based HSI super-resolution and single HSI super-resolution. The fusion-based HSI super-resolution usually applies a high spatial resolution multispectral image as the auxiliary information to assist the low spatial resolution hyperspectral image to obtain a high spatial resolution hyperspectral image. Many methods such as matrix or tensor factorization [3], [4], [5], sparse representation [6], and deep learning [7], [8], [9], [10], [11] have been proposed in recent years and also achieved considerable performance. However, these approaches usually require a good registration processing between the HR auxiliary image and the observed low resolution hyperspectral image (LR-HSI), which is also challenging in practical applications.

The single HSI super-resolution technique overcomes the limitation mentioned above, which infers the desired high resolution hyperspectral image (HR-HSI) only by its corresponding observed LR-HSI, without requiring any auxiliary images. Therefore, this technique has attracted more and more attention in recent years. In the past decades, the traditional single super-resolution techniques have been successfully applied into the gray/red green blue (RGB) color image super-resolution. Generally speaking, these methods are further divided into learning-based methods and model-based methods. The learning-based method usually applies the learning models, such as convolutional neural network (CNN) [12] and dictionary learning [13], to learn the mapping relationships between the LR images and their corresponding HR images from millions of training samples. The

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model-based method usually applies different priors, such as sparse prior, nonlocal priors [14], [15], and autoregressive model [16], of the inner structure to establish different regularization terms to the reconstruction model. Although these traditional single super-resolution techniques have obtained reliable performance for natural images, it is very difficult to directly copy them into the field of single HSI super-resolution because of the fact that bandwise reconstruction will lose the high correlation of spectral bands and result in spectral distortion [17] in the super-resolved HSI [18], [19], [20]. Consequently, recent researches pay more attention to inherent spatial–spectral correlation preservation.

1) *CNN-Based Single HSI Super-Resolution:* First, we analyze the learning-based single HSI super-resolution. In recent years, the CNN with its strong learning ability has already been demonstrated to be a very feasible way for HSI tasks [21], [22], [23]. As for the CNN-based single HSI super-resolution, it usually applies the spectral difference learning or 3-D convolution to enable spatial–spectral correlation preservation. Li and Hu proposed three HSI super-resolution networks named spatial constraint (SCT)_spectral difference CNN (SDCNN) [24], spatial error-correction (SEC)_SDCNN [25], and intrafusion network (IFN) [26], all of which apply spectral difference learning to maintain the correlation among different spectral bands. Hu proposed an HSI super-resolution method by combining an information distillation network with an intrafusion operation [27], which applies an intrafusion strategy to capture the correlation between the selected bands and the unselected bands. Mei *et al.* [17] proposed 3D full CNN (3DFCNN) for single HSI super-resolution, which adopts the 3-D all CNN (3DFCNN) to leverage the spatial and spectral correlation of the neighborhood bands. Jiang *et al.* [28] proposed a single HSI super-resolution method named spatial–spectral prior super-resolution (SSPSR), which designs a spectral attention residual module to capture the correlation among different bands. Moreover, the newly reconstructed spectral band is formed by weighting all the spectral bands. However, the above methods also have some limitations. The first is that 3-D convolution needs a large number of parameters, which will inevitably lead to large time complexity. Although many literature (exploring the relationship between 2D/3D convolution super-resolution (ERCSR) [29], mixed 2D/3D convolutional network (MCNet) [30], and spectrum and feature context super-resolution (SFCSR) [31]) applied separable 3-D convolution and other literature (SSPSR [28], recurrent feedback super-resolution (RFSR) [32], grouped deep recursive residual network (GDRRN) [33], and spectral grouping and attention-driven residual dense network (SGARDN) [34]) applied the spectral bands' grouping strategy to reduce the number of parameters, the time complexity is still large. The second is that the CNN-based methods belong to the supervised learning method, which needs a large number of training samples to train the network model. However, many HSIs may be obtained using different sensors or cameras, and hence, the available training HSIs are extremely small, which is hard to guarantee the training process [28], [35]. Although many studies consider augmenting the training images by flipping horizontally or vertically, this strategy cannot generate new image structures. The last is that different HSIs acquired using different HSI sensors need to train their network models independently, which will inevitably increase the training burden.

2) *Model-Based Single HSI Super-Resolution:* Second, we analyze the model-based single HSI super-resolution. The model-based methods usually apply the matrix factorization strategy and the prior strategy to capture the spatial–spectral correlation. As for the matrix factorization strategy, researchers usually rearrange the desired HR-HSI with a 3-D form into a spectral or spatial matrix with a 2-D form and assume that the matrix can be mapped into a low-dimensional subspace using a spatial or spectral dictionary and a set of spatial or spectral coefficients. The spatial or spectral correlation can be captured by performing a variety of priors on the spatial or spectral coefficients. Zhao applied the spatial sparsity and spectral regularizations to maintain the spatial–spectral consistency [36]. Li proposed an HSI super-resolution method using the spectral mixture analysis and spatial–spectral group sparsity [37], which applies the nonlocal self-similarity and the sparsity to capture the spatial–spectral consistency. This strategy has been widely used in the fusion-based HSI super-resolution methods [6], [38], [39]; however, it is rare in single HSI super-resolution. As for the prior strategy, researchers usually directly integrate different priors into the desired HR-HSI. Shi proposed low-rank and total variation (LRTV) [40], which applies the low-rank and total variation (TV) regularizations to reconstruct a high-resolution magnetic resonance (MR) image. Wang proposed nonlocal low-rank tensor approximation and total variation (NLRTATV) [19], which combines low-rank tensor approximation and TV regularization to capture the spatial–spectral correlation. Li combined low tensor train rank representation and 3-D TV regularization to capture the spatial–spectral consistency of the reconstructed HSIs [41]. He [42] also applied 3-D TV regularization and tensor low rank to capture the spatial–spectral consistency of reconstructed HSIs. Since the model-based method has no training phase, it can overcome the second limitation of the CNN-based method (available training HSIs are extremely small). However, it is a double-edged sword because no training phase will lose the reconstruction accuracy. In addition, the spectral or spatial matrix factorization methods also have another limitation that different scenes are inaccurate represented using only the same single spectral or spatial dictionary.

A. Motivation and Contribution

According to the above analyses, we can conclude that the model-based and learning-based methods are complementary in the single HSI super-resolution task, especially under

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the small study sample condition. As discussed earlier, the model-based method does not need training samples, but loses the reconstruction accuracy. Conversely, the learning-based method can achieve well reconstruction accuracy with massive training samples; however, the available training HSIs are extremely few [28]. This motivates us to explore two issues: 1) how to design a mapping relationship between the training LR-HSI and HR-HSI pairs with relatively few samples and 2) how to combine the advantages of the learning-based and model-based methods. It is well-known that that CNN-based methods usually require a large number of samples to train massive convolutions and the trained network is not versatile for different HSIs. Therefore, we apply dictionary learning to explore this mapping relationship because it contains relatively less parameters than the CNN and it can learn the microstructure [14] relationship using relatively small patches. Consequently, it needs less training samples than the CNN-based method. Moreover, we can easily combine dictionary learning with some existing priors to create more robust super-resolution model.

With all the above analyses in mind, this article presents a novel single HSI super-resolution framework by combining a grouped joint tensor dictionary (GJTD) with a low-rank prior (termed as GJTD-LR). GJTD-LR contains two main stages. The first stage is GJTD learning, which is the core contribution in this article. In GJTD learning, we design a novel mapping relationship between training HR-HSIs and LR-HSIs using the tensor–tensor product (t-product) [43]. Specifically, we first divide the training HR-HSIs and LR-HSIs into several groups and learn a compact joint tensor subdictionary for each group. To improve the robustness of dictionary representation, the training HSIs in each group can not only be strongly represented by its compact joint tensor subdictionary but also be weakly represented by other joint tensor subdictionaries. Moreover, GJTD can preserve the correlation among different spectral bands due to the characteristic of t-product. Since this is a supervised learning stage, the training information can be used. The grouped strategy can overcome the limitation in the matrix factorization methods (different scenes using only the same single dictionary). The second stage is the reconstruction model. Specifically, we first get a preoptimization value of the tensor sparse coefficient using the acquired LR-HSI over the LR dictionary, and then design a newly reconstruction model by integrating the HR dictionary, preoptimization tensor sparse coefficient, and the low-rank prior to get the desired HR-HSI. Since it is an unsupervised learning stage, the problem of inadequately training HSIs can be alleviated.

To the best of our knowledge, it is the first time to use the grouped strategy, the t-product, and the joint dictionary learning into the single HSI super-resolution problem. Although GJTD-LR belongs to the model-based methods, it is different from them. Unlike the model-based single HSI super-resolution, which has no learning phase and usually applies different priors to create the reconstruction model, GJTD-LR not only applies a tensor low-rank prior to constrain the desired HR-HSI but also creates a novel mapping relationship between the training HR-HSIs and LR-HSIs to better get the desired HR-HSI. GJTD-LR can also be regarded as a learning-based method; however, it is also different from the CNN-based methods because of their learning mechanisms. Unlike the original joint dictionary [13], which learns a single joint dictionary for 2-D training samples, GJTD learns multiple sub-dictionaries for 3-D training samples using t-product. Hence, GJTD can capture the correlation among different spectral bands.

In summary, the main contributions of this article are listed as follows.

1) Aiming at insufficient training samples of HSIs, we proposed a novel single HSI super-resolution method GJTD-LR, which combines a GJTD learning model and a low-rank prior to get a high-quality HR-HSI from an observed LR-HSI.
2) To learn an accurate mapping relationship between a pair of training LR-HSI and HR-HSI under small sample conditions, we innovatively proposed a GJTD by combining the grouped strategy and t-product. The grouped strategy provides two types of dictionaries for each training sample and the t-product preserves the intrinsic spectral correlation among different spectral bands.
3) To optimize the proposed model, we designed a novel optimized algorithm based on alternating direction method of multipliers (ADMM), soft threshold algorithm, singular value decomposition (SVD), and Fourier domain transform.
4) The experimental results confirm that GJTD-LR clearly improves the reconstruction results compared with some of the new techniques in both the quantitative evaluation and visual comparison.

The rest of this article is structured as follows. In Section II, we will restate the related works about the tensor notation and t-product. The proposed GJTD-LR and its corresponding optimization process are introduced in Section III. Section IV illustrates the experimental results and parameters' discussion. Finally, we conclude this article in Section V.

II. Related Work

This section provides some fundamental properties used in GJTD-LR, including the basic tensor notations and the t-product model. Since they have been introduced in many literature, we also restate them here for the convenience of reading.

A. Tensor Notations

Similar to many notations in the existing literature [44], higher order tensors are denoted by calligraphic letters, e.g., $\mathcal{A}$, matrices are denoted by capital letters, e.g., A. For a third-order tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $\mathcal{A}(i_1, i_2, i_3)$ denotes the element value of $\mathcal{A}$ in location (i_1, i_2, i_3), where $i_1 \in \{1, \ldots, n_1\}$, $i_2 \in \{1, \ldots, n_2\}$ and $i_3 \in \{1, \ldots, n_3\}$. $\mathcal{A}(i_1, :, :)$, $\mathcal{A}(:, i_2, :)$, and $\mathcal{A}(:, :, i_3)$ denote the horizontal, lateral, and frontal slices of $\mathcal{A}$, respectively. $\mathcal{A}(:, i_2, i_3)$, $\mathcal{A}(i_1, :, i_3)$, and $\mathcal{A}(i_1, i_2, :)$ denote the mode-1, mode-2, and mode-3 fibers of $\mathcal{A}$, respectively. The mode-3 unfolding is denoted as $A_{(3)} = \text{unfold}_3(\mathcal{A}) \in R^{n_3 \times n_1 n_2}$ by arranging the mode-3 fibers as columns. The mode-3 folding is the inverse operation of mode-3 unfolding, which denotes $\mathcal{A} = \text{fold}_3(A_{(3)})$. The tensor l_1 norm is defined as $\|\mathcal{A}\|_1 = \sum_{i_1=1}^{n_1} \sum_{i_2=1}^{n_2} \sum_{i_3=1}^{n_3} |\mathcal{A}(i_1, i_2, i_3)|$, and the tensor Frobenius norm is defined as $\|\mathcal{A}\|_F = (\sum_{i_1=1}^{n_1} \sum_{i_2=1}^{n_2} \sum_{i_3=1}^{n_3} \mathcal{A}(i_1, i_2, i_3)^2)^{1/2}$.

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B. Tensor-Tensor Product

Recently, t-product is widely used in the HSI processing tasks such as fuse-based HSI super-resolution [45] and hyperspectral (HS) anomaly detection [46]. Since discrete Fourier transformation (DFT) [47] plays an important role in t-product, we introduce it first. For a vector $x \in \mathbb{R}^n$, the transformed vector $\overline{x} \in \mathbb{C}^n$ is obtained using $\overline{x} = F_n x$, where $F_n \in \mathbb{C}^{n\times n}$ is the 1-D DFT. Based on this, $\overline{\mathcal{A}} \in \mathbb{C}^{n_1\times n_2\times n_3}$ denotes the transformed tensor by performing 1-D DFT along the mode-3 fibers of $\mathcal{A}$. In fact, $\overline{\mathcal{A}}$ can be computed directly by the MATLAB command $\overline{\mathcal{A}} = \text{fft}(\mathcal{A}, [\,], 3)$, and the inverse operation can also be computed by $\mathcal{A} = i\text{fft}(\overline{\mathcal{A}}, [\,], 3)$. $\overline{\mathcal{A}}$ can be transformed into a block diagonal matrix $\overline{A} \in \mathbb{C}^{n_1n_3\times n_2n_3}$ by

$$\overline{A} = \text{bdiag}\left(\overline{\mathcal{A}}\right) = \begin{bmatrix} \overline{A}^{(1)} & & & \\ & \overline{A}^{(2)} & & \\ & & \ddots & \\ & & & \overline{A}^{(n_3)} \end{bmatrix} \tag{1}$$

where bdiag$(\cdot)$ represents the operator that maps the tensor $\overline{\mathcal{A}}$ to the block diagonal matrix $\overline{A}$. $\overline{A}^{(i_3)}$ is the i_3th frontal slide of $\overline{\mathcal{A}}$.

Then the block circulant matrix bcirc$(\mathcal{A}) \in R^{n_1n_3\times n_2n_3}$ of $\mathcal{A}$ is defined as

$$\text{bcirc}(\mathcal{A}) = \begin{bmatrix} A^{(1)} & A^{(n_3)} & \cdots & A^{(2)} \\ A^{(2)} & A^{(1)} & \cdots & A^{(3)} \\ \vdots & \vdots & \vdots & \vdots \\ A^{(n_3)} & A^{(n_3-1)} & \cdots & A^{(1)} \end{bmatrix}. \tag{2}$$

Using the unfold operator, $\mathcal{A}$ is transformed into a matrix of size $n_1n_3 \times n_2$. fold is its inverse operator

$$\text{unfold}(\mathcal{A}) = \begin{bmatrix} A^{(1)} \\ A^{(2)} \\ \vdots \\ A^{(n_3)} \end{bmatrix}, \quad \text{fold}(\text{unfold}(\mathcal{A})) = \mathcal{A}. \tag{3}$$

Definition 1 (t-product): Let $\mathcal{A} \in \mathbb{R}^{n_1\times n_2\times n_3}$ and $\mathcal{B} \in \mathbb{R}^{n_2\times n_4\times n_3}$, the t-product of them is defined as a tensor $\mathcal{C} \in R^{n_1\times n_4\times n_3}$

$$C = \mathcal{A} * \mathcal{B} = \text{fold}(\text{bcirc}(\mathcal{A}) \cdot \text{unfold}(\mathcal{B})). \tag{4}$$

To simplify the calculation, it is an efficient way to compute t-product [43] based on fast Fourier transform (FFT) as shown in Algorithm 1.

Algorithm 1 Tensor–Tensor Product

Input: $\mathcal{A} \in \mathbb{R}^{n_1\times n_2\times n_3}, \mathcal{B} \in \mathbb{R}^{n_2\times n_4\times n_3}$

Output: $\mathcal{C} = \mathcal{A} * \mathcal{B} \in \mathbb{R}^{n_1\times n_4\times n_3}$

1 Compute $\overline{\mathcal{A}} = fft(\mathcal{A}, [\,], 3)$ and $\overline{\mathcal{B}} = fft(\mathcal{B}, [\,], 3)$;

2 Compute each frontal slice of $\overline{\mathcal{C}}$ by

$$\overline{\mathcal{C}}^{(i_3)} = \begin{cases} \bar{A}^{(i_3)}\bar{B}^{(i_3)} & i_3 = 1, \ldots, \lceil \frac{n_3+1}{2} \rceil \\ conj(\bar{\mathcal{C}}^{(n_3-i_3+2)}) & i_3 = \lceil \frac{n_3+1}{2} \rceil + 1, \ldots, n_3; \end{cases}$$

3 Compute $\mathcal{C} = ifft(\bar{\mathcal{C}}, [\,], 3)$.

There are two properties which will be used in the following.

1) $\|\mathcal{C}\|_F^2 = (1/n_3)\|\overline{C}\|_F^2$,
2) $\mathcal{C} = \mathcal{A} * \mathcal{B}$ is equal to $\overline{C} = \overline{AB}$ and $\overline{C}^{(i_3)} = \overline{A}^{(i_3)}\overline{B}^{(i_3)}$, where $i_3 \in \{1, \ldots, n_3\}$.

More details are referred to [48], [49].

III. Proposed GJTD-LR Approach

In this section, we will introduce the procedure of the proposed GJTD-LR, the flowchart of which is shown in Fig. 1.

A. GJTD Model

Learning a better mapping relationship between the training LR-HSIs and HR-HSIs is crucial to single image super-resolution. Widely used mapping relationships include the joint dictionary learning and the CNN. The characteristic of the CNN has been analyzed above. For the joint dictionary learning, which is proposed in literature [13], it applies an LR and HR dictionary pair to build the mapping relationship between the LR images and their corresponding HR images, and it achieves success in natural image super-resolution. However, it is difficult to be directly applied to HSI super-resolution, mainly because of the fact that the joint dictionary [13] usually loses the spectral correlation of HSIs.

According to the definition of the t-product (as shown in Algorithm 1), $\mathcal{C}(i, j, :)$ represents the (i, j)th mode-3 fiber of $\mathcal{C}$ and it is obtained by $\mathcal{C}(i, j, :) = \sum_l^{n_2} \mathcal{A}(i, l, :) * \mathcal{B}(l, j, :)$. Repeating i from 1 to n_1, the (i, j)th mode-3 fiber $\mathcal{C}(i, j, :)$ is extended into the jth lateral slice $\mathcal{C}(:, j, :)$. Based on these analyses and literature [45], $\mathcal{C}(:, j, :)$ is defined as in (5)

$$\mathcal{C}(:, j, :) = \mathcal{A} * \mathcal{B}(:, j, :) = \sum_l^{n_2} \mathcal{A}(:, l, :) * \mathcal{B}(l, j, :). \tag{5}$$

Equation (5) is an interesting inference because the jth lateral slice of $\mathcal{C}$, i.e., $\mathcal{C}(:, j, :) \in \mathbb{R}^{n_1\times 1\times n_3}$, is decomposed into a tensor dictionary $\mathcal{A} \in \mathbb{R}^{n_1\times n_2\times n_3}$ and its corresponding tensor coefficient $\mathcal{B}(:, j, :) \in \mathbb{R}^{n_1\times 1\times n_3}$. Supposing that $\mathcal{C}(:, j, :)$ represents n_1 pixels (arranged by column) with n_3 bands in an HSI, the spectral correlation is preserved by constraining the tensor sparse coefficient $\mathcal{B}(:, j, :)$.

In addition, the traditional dictionary representation also has another limitation. Early studies usually learn a universal dictionary for different training scenes, which will obtain a noncompact dictionary. Afterward, the following studies overcome this limitation by learning different dictionaries for different training scenes [14]. However, it is still not very perfect because each training scene is independent of its corresponding dictionary. The Fisher discrimination dictionary [50] provided a novel scheme to pattern classification, which applies a structured dictionary to represent the training images. In this scheme, the samples in each group are represented not only by its corresponding dictionary but also by the whole dictionary.

Motivation comes from the joint dictionary, structured dictionary, and t-product, and we propose a novel GJTD model, which contains two phases *tensor grouped transform* and *modeling* based on t-product.

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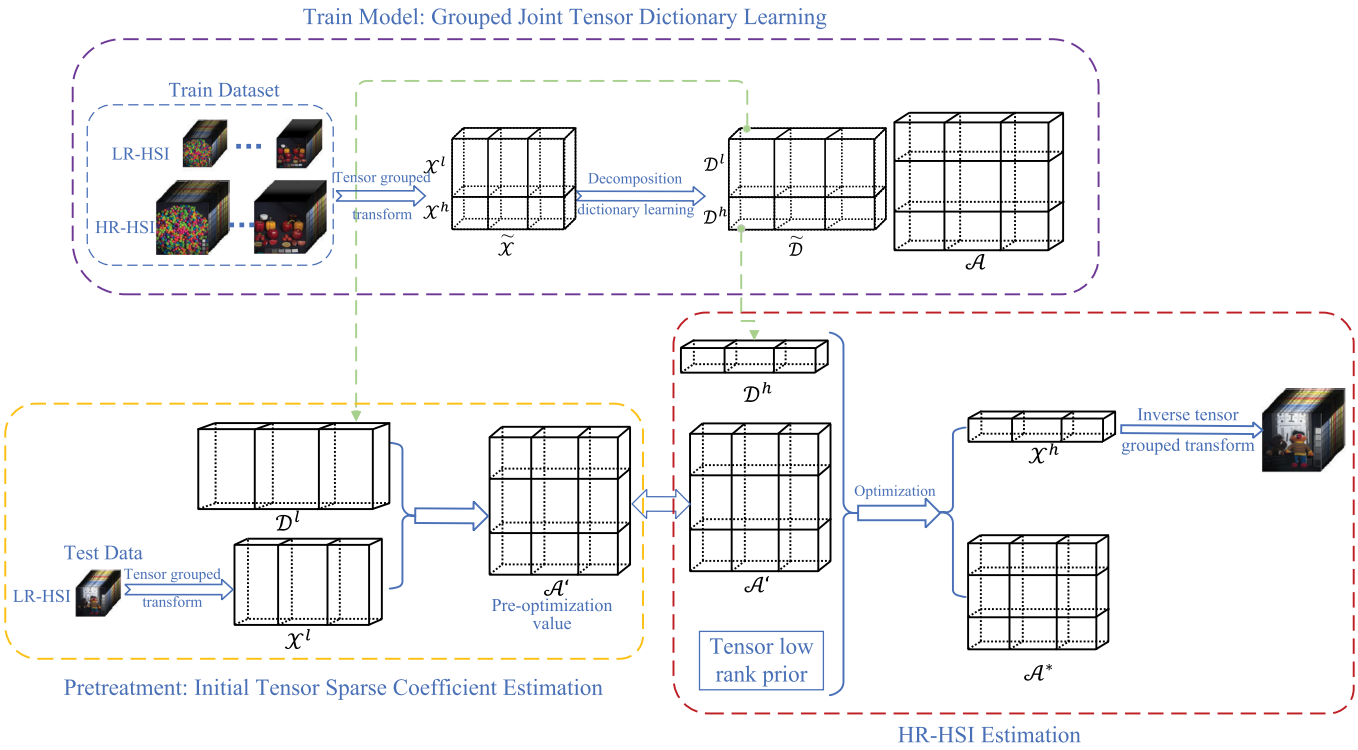


Fig. 1. Flowchart of the proposed GJTD-LR.

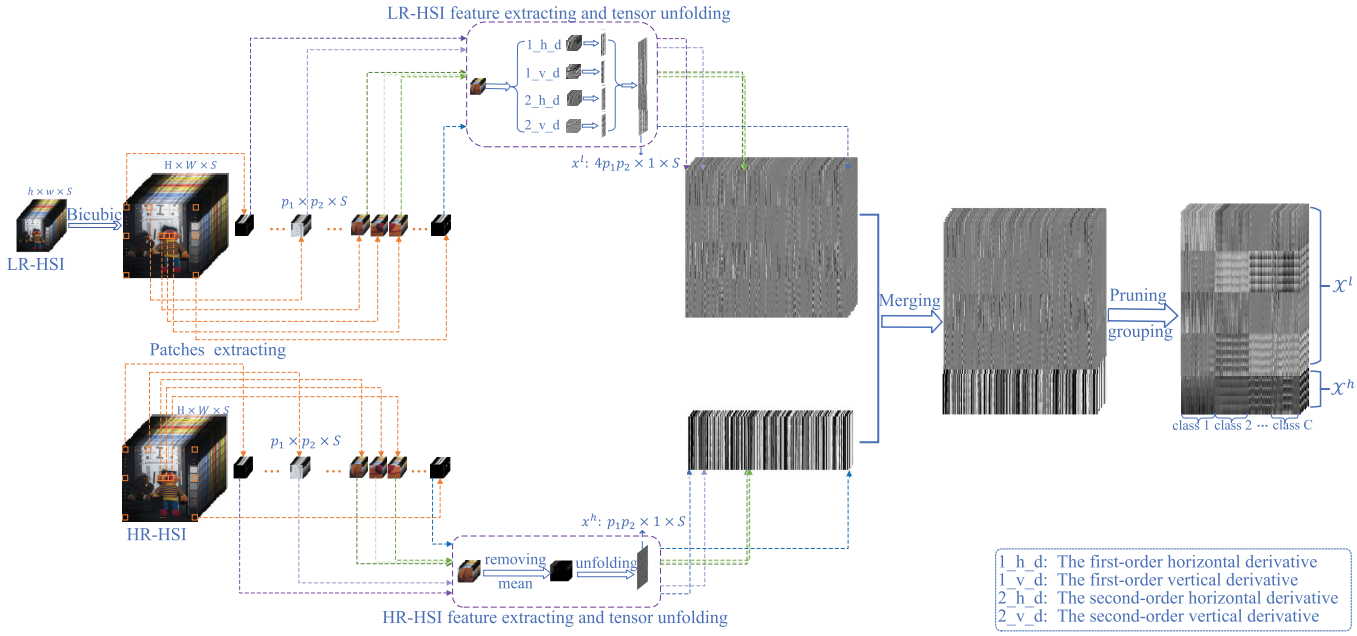


Fig. 2. Procedure of the *tensor grouped transform*.

First, we introduce the *tensor grouped transform*, the details of which are shown in Fig. 2. Before this operation, each LR-HSI of size $w \times h \times S$ is interpolated into the same size as HR-HSI, i.e., $W \times H \times S$, where w, h, and S are the dimensions of the width, height, and spectral bands, respectively, in LR-HSI. W and H are the dimensions of the width and height in HR-HSI, respectively. They satisfy $W = SF \times w$ and $H = SF \times h$. SF is the scaling factor. *Tensor grouped transform* is used to transform LR-HSI and HR-HSI into the corresponding grouped tensor forms $\mathcal{X}^l$ and $\mathcal{X}^h$, respectively. First, LR-HSIs and HR-HSIs are divided into n overlapping 3-D patches of size $p_1 \times p_2 \times S$ to build two 3-D patch sets, respectively. Second, the features of each 3-D patch are extracted band by band. We apply the first-order and second-order derivatives in the horizontal and vertical directions to extract the features of LR-HSI patch as suggested

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in [13]. For HR-HSI patch, we remove the mean in each band to form the zero-centered feature. Hence, we can get four feature cubes for each LR-HSI patch and a single feature cube for each HR-HSI. Third, each LR-HSI feature cube is unfolded to form a 2-D matrix by mode-3 unfolding operation. Four feature cubes of each LR-HSI patch are concatenated into a matrix $x^l \in \mathbb{R}^{4p_1p_2\times S}$, and the single feature cube of each HR-HSI patch is also unfolded into a matrix $x^h \in \mathbb{R}^{p_1p_2\times S}$. In the following, we use $P = 4p_1p_2$ and $\tilde{P} = p_1p_2$ to simplify the description. The new 2-D matrices are seen as two latest slices of $\mathcal{X}^l$ and $\mathcal{X}^h$, respectively, such as $x^l \in \mathbb{R}^{P\times 1\times S}$ and $x^h \in \mathbb{R}^{\tilde{P}\times 1\times S}$. Fourth, each LR-HSI latest slice with its corresponding HR-HSI latest slice are merged to form a joint latest slice $\tilde{x} \in \mathbb{R}^{(P+\tilde{P})\times 1\times S}$. Fifth, we use the pruning operator to preserve the meaningful joint latest slices with a certain amount of edge structures [14]. Specifically, we calculate the intensity variance of each joint latest slice, discarding the joint latest slice with small intensity variance and preserving the joint latest slice with large intensity variance. Finally, we use the grouping operator to divide the remaining joint latest slices into $C \in \mathbb{R}$ groups using the K-means clustering algorithm. The grouped tensor is formed as $\mathcal{X}^l = [\mathcal{X}_1^l, \ldots, \mathcal{X}_C^l]$ and $\mathcal{X}^h = [\mathcal{X}_1^h, \ldots, \mathcal{X}_C^h]$. C_i represents the number of the ith group and it satisfies $\sum_{i=1}^{C} C_i = n$ and $i \in \{1, \ldots, C\}$. n represents the total number of latest slices. $\mathcal{X}_i^l \in \mathbb{R}^{P\times C_i\times S}$ denotes the ith group.

Second, we introduce *modeling* based on t-product. According to the idea of the joint dictionary, we need to learn a pair of LR dictionary $\mathcal{D}^l$ and HR dictionary $\mathcal{D}^h$. In addition, according to the idea of the structured dictionary, each group should learn a corresponding compact dictionary. Therefore, the LR dictionary and HR dictionary can be represented as $\mathcal{D}^l = [d_1^l, \ldots, d_m^l] = [\mathcal{D}_1^l, \ldots, \mathcal{D}_C^l] \in \mathbb{R}^{P\times m\times S}$ and $\mathcal{D}^h = [d_1^h, \ldots, d_m^h] = [\mathcal{D}_1^h, \ldots, \mathcal{D}_C^h] \in \mathbb{R}^{\tilde{P}\times m\times S}$, respectively. $d_j^l \in \mathbb{R}^{P\times 1\times S}$ and $d_j^h \in \mathbb{R}^{\tilde{P}\times 1\times S}$ represent the jth atoms in $\mathcal{D}^l$ and $\mathcal{D}^h$, respectively. m represents the number of atoms of the whole dictionary. $\mathcal{D}_i^l \in \mathbb{R}^{P\times k_i\times S}$ and $\mathcal{D}_i^h \in \mathbb{R}^{\tilde{P}\times k_i\times S}$ represent the HR subdictionary and LR subdictionary corresponding to the ith group, respectively. k_i is the number of atoms of the ith subdictionary and it satisfies $\sum_{i=1}^{C} k_i = m$. In general, we set the same number of autos in each subdictionary, i.e., $k_i = m/C$. For each 3-D patch, we can get a tensor sparse coefficient, and for each group, we can get a set of tensor sparse coefficients and the whole tensor sparse coefficients can be expressed as $\mathcal{A} = [a_1, \ldots, a_n] = [\mathcal{A}_1, \ldots, \mathcal{A}_C] \in \mathbb{R}^{m\times n\times S}$. $a_l \in \mathbb{R}^{m\times 1\times S}$ and $A_i \in \mathbb{R}^{m\times C_i\times S}$ represent the lth tensor sparse coefficient and the ith group tensor sparse coefficients, respectively, where $l \in \{1, \ldots, n\}$. Based on the above analyses, integrating the ideas of the t-product, the structured dictionary, and the joint dictionary, we propose the GJTD model to learn the LR and HR dictionary pairs as shown in (6)

$$\begin{aligned}\underset{(\mathcal{D}^h,\mathcal{D}^l,\mathcal{A})}{\arg\min}\ & \frac{1}{P}\left(\sum_{i=1}^{C}\left(\left\|\mathcal{X}_i^h-\mathcal{D}^h*\mathcal{A}_i\right\|_F^2+\left\|\mathcal{X}_i^h-\mathcal{D}_i^h*\mathcal{A}_i^i\right\|_F^2\right.\right.\\ &\left.\left.+\sum_{j=1,j\neq i}^{C}\left\|\mathcal{D}_j^h*\mathcal{A}_i^j\right\|_F^2\right)\right)\\ &+\frac{1}{\tilde{P}}\left(\sum_{i=1}^{C}\left(\left\|\mathcal{X}_i^l-\mathcal{D}^l*\mathcal{A}_i\right\|_F^2+\left\|\mathcal{X}_i^l-\mathcal{D}_i^l*\mathcal{A}_i^i\right\|_F^2\right.\right.\\ &\left.\left.+\sum_{j=1,j\neq i}^{C}\left\|\mathcal{D}_j^l*\mathcal{A}_i^j\right\|_F^2\right)\right)\\ &+\tilde{\lambda}\left(\frac{1}{P}+\frac{1}{\tilde{P}}\right)\|\mathcal{A}\|_1 \qquad (6)\end{aligned}$$

where $\tilde{\lambda}$ is used to balance the tradeoff between the dictionary representation error and sparsity. We can also simplify this model to

$$\begin{aligned}\underset{\tilde{\mathcal{D}},\mathcal{A}}{\arg\min}\ & \sum_{i=1}^{C}\left(\left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}*\mathcal{A}_i\right\|_F^2+\left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_i*\mathcal{A}_i^i\right\|_F^2\right.\\ &\left.+\sum_{j=1,j\neq i}^{C}\left\|\tilde{\mathcal{D}}_j*\mathcal{A}_i^j\right\|_F^2\right)+\mu\|\mathcal{A}\|_1 \qquad (7)\end{aligned}$$

where $\tilde{\mathcal{X}} = \begin{bmatrix}(1/\sqrt{P})\mathcal{X}_i^l\\(1/\sqrt{\tilde{P}})\mathcal{X}_i^h\end{bmatrix}$, $\tilde{\mathcal{D}} = \begin{bmatrix}(1/\sqrt{P})\mathcal{D}^l\\(1/\sqrt{\tilde{P}})\mathcal{D}^h\end{bmatrix}$ and $\mu = \tilde{\lambda}((1/P)+(1/\tilde{P}))$.

The first and second terms imply that the patches in each group not only can be strongly represented by their corresponding subdictionary but also can be weakly represented by other subdictionaries. It can increase the robustness of dictionary representation for different structural samples.

B. GJTD Model Optimization

It is clear that, (7) is a nonconvex optimization problem. However, it is converted into a convex optimization problem when optimizing one variable while fixing other variables. For efficiency, we adopt the splitting strategy and the ADMM [51] to solve it. The above biconvex optimization is efficiently solved by solving the following two subproblems iteratively:

$$\begin{cases}\tilde{\mathcal{D}}^{\text{iter}+1} = \underset{\tilde{\mathcal{D}}}{\arg\min}\, L\left(\mathcal{A}^{\text{iter}}, \tilde{\mathcal{D}}\right)\\ \mathcal{A}^{\text{iter}+1} = \underset{\mathcal{A}}{\arg\min}\, L\left(\mathcal{A}, \tilde{\mathcal{D}}^{\text{iter}}\right)\end{cases} \qquad (8)$$

where $L(\cdot)$ and iter denote the objective function and the iteration number, respectively.

1) Optimization With Respect to $\tilde{\mathcal{D}}$: With a fixed tensor sparse coefficient $\mathcal{A}$, we optimize the dictionary $\tilde{\mathcal{D}}$ according to the first term of (8). This model is rewritten as

$$\begin{aligned}\underset{\tilde{\mathcal{D}}}{\arg\min}\ & \sum_{i=1}^{C}\left(\left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}*\mathcal{A}_i\right\|_F^2+\left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_i*\mathcal{A}_i^i\right\|_F^2\right.\\ &\left.+\sum_{j=1,j\neq i}^{C}\left\|\tilde{\mathcal{D}}_j*\mathcal{A}_i^j\right\|_F^2\right). \qquad (9)\end{aligned}$$

Since each $\tilde{\mathcal{D}}_i$ is independent, we can optimize it one by one. The above model is converted to the following:

$$\begin{aligned}\underset{\tilde{\mathcal{D}}_i}{\arg\min}\ & \left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_i*\mathcal{A}_i^i-\sum_{j=1,j\neq i}^{C}\tilde{\mathcal{D}}_j*\mathcal{A}_i^j\right\|_F^2\\ &+\left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_i*\mathcal{A}_i^i\right\|_F^2. \qquad (10)\end{aligned}$$

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In here, we apply the Fourier domain method, which is a widely used method in many literature [45], [48], [49], to solve (10). We convert (10) into Fourier domain as

$$\arg\min_{\overline{\tilde{D}_i}} \frac{1}{S}\left\|\overline{\tilde{X}_i}-\overline{\tilde{D}_i A_i^i}-\sum_{j=1,j\neq i}^{C}\overline{\tilde{D}_j A_i^j}\right\|_F^2 + \frac{1}{S}\left\|\overline{\tilde{X}_i}-\overline{\tilde{D}_i A_i^i}\right\|_F^2 \quad (11)$$

which is equal to

$$\arg\min_{\overline{\tilde{D}_i}} \sum_{s=1}^{S}\left\|\overline{\tilde{X}_i}^{(s)}-\overline{\tilde{D}_i}^{(s)}\overline{A_i^i}^{(s)}-\sum_{j=1,j\neq i}^{C}\overline{\tilde{D}_j}^{(s)}\overline{A_i^j}^{(s)}\right\|_F^2 + \sum_{s=1}^{S}\left\|\overline{\tilde{X}_i}^{(s)}-\overline{\tilde{D}_i A_i^i}^{(s)}\right\|_F^2 \quad (12)$$

where s represents the sth frontal slide. The optimal $\overline{\tilde{D}_i}^{(s)}$ is

$$\overline{\tilde{D}_i}^{(s)} = \left(\overline{\tilde{X}_i}^{(s)}\overline{A_i^i}^{(s)T} - \frac{1}{2}\sum_{j=1,j\neq i}^{C}\overline{\tilde{D}_j}^{(s)}\overline{A_i^j}^{(s)}\overline{A_i^i}^{(s)T}\right) \times \left(\overline{A_i^i}^{(s)}\overline{A_i^i}^{(s)T}\right)^{-1}. \quad (13)$$

Then the optimal $\tilde{\mathcal{D}}_i$ is computed using $\tilde{\mathcal{D}}_i = ifft(\overline{\tilde{D}_i}, [\,], 3)$.

2) Optimization With Respect to $\mathcal{A}$: With a fixed dictionary $\tilde{\mathcal{D}}$, we optimize the tensor sparse coefficient $\mathcal{A}$ according to the second equation of (8). Similar to the optimization of $\tilde{\mathcal{D}}_i$, we also decompose this model into several subproblems. For optimizing $\mathcal{A}_i$, this model is rewritten as

$$\arg\min_{\mathcal{A}_i} \left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}*\mathcal{A}_i\right\|_F^2 + \left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_i*\mathcal{A}_i^i\right\|_F^2 + \sum_{j=1,j\neq i}^{C}\left\|\tilde{\mathcal{D}}_j*\mathcal{A}_i^j\right\|_F^2 + \mu\|\mathcal{A}_i\|_1. \quad (14)$$

Since $\mathcal{A}_i$ is decomposed into two parts, i.e., $\mathcal{A}_i^i$ and $\mathcal{A}_i^j$, the model can be rewritten again to

$$\arg\min_{\mathcal{A}_i^i,\mathcal{A}_i^j} \left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_i*\mathcal{A}_i^i-\sum_{j=1,j\neq i}^{C}\tilde{\mathcal{D}}_j*\mathcal{A}_i^j\right\|_F^2 + \left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_i*\mathcal{A}_i^i\right\|_F^2 + \sum_{j=1,j\neq i}^{C}\left\|\tilde{\mathcal{D}}_j*\mathcal{A}_i^j\right\|_F^2 + \mu\left\|\mathcal{A}_i^i\right\|_1 + \mu\sum_{j=1,j\neq i}^{C}\left\|\mathcal{A}_i^j\right\|_1. \quad (15)$$

a) Optimization of $\mathcal{A}_i^i$: Since optimizing $\mathcal{A}_i^i$ needs to fix $\mathcal{A}_i^j$ with $j = 1,\ldots,C, j\neq i$, we simplify the model using $\mathcal{H}_1 = \tilde{\mathcal{X}}_i - \sum_{j=1,j\neq i}^{C}\tilde{\mathcal{D}}_j*\mathcal{A}_i^j$

$$\arg\min_{\mathcal{A}_i^i} \left\|\mathcal{H}_1-\tilde{\mathcal{D}}_i*\mathcal{A}_i^i\right\|_F^2 + \left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_i*\mathcal{A}_i^i\right\|_F^2 + \mu\left\|\mathcal{A}_i^i\right\|_1. \quad (16)$$

We apply the splitting strategy to minimize each term in (16) separately, instead of the sum of all the terms. Equation (16) is rewritten as

$$\arg\min_{\mathcal{A}_i^i} \left\|\mathcal{H}_1-\tilde{\mathcal{D}}_i*\mathcal{B}_1\right\|_F^2 + \left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_i*\mathcal{B}_1\right\|_F^2 + \mu\left\|\mathcal{A}_i^i\right\|_1 \quad \text{s.t. } \mathcal{B}_1 = \mathcal{A}_i^i. \quad (17)$$

$\mathcal{B}_1 = \mathcal{A}_i^i$ is the splitting variable. Then, the augmented Lagrangian function of (17) becomes

$$\arg\min_{\mathcal{A}_i^i,\mathcal{B}_1} \left\|\mathcal{H}_1-\tilde{\mathcal{D}}_i*\mathcal{B}_1\right\|_F^2 + \left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_i*\mathcal{B}_1\right\|_F^2 + \mu_1\left\|\mathcal{A}_i^i-\mathcal{B}_1+\frac{\Lambda_1}{2\mu_1}\right\|_F^2 + \mu\left\|\mathcal{A}_i^i\right\|_1 \quad (18)$$

where $\Lambda_1 \in \mathbb{R}^{k_i\times C_i\times S}$ denotes the Lagrangian multiplier ($\mu_1 > 0$). Equation (18) is solved by the following two subproblems iteratively.

$\mathcal{B}_1$ Subproblem: When optimizing $\mathcal{B}_1$, the model is written as

$$\arg\min_{\mathcal{B}_1} \left\|\mathcal{H}_1-\tilde{\mathcal{D}}_i*\mathcal{B}_1\right\|_F^2 + \left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_i*\mathcal{B}_1\right\|_F^2 + \mu_1\left\|\mathcal{A}_i^i-\mathcal{B}_1+\frac{\Lambda_1}{2\mu_1}\right\|_F^2. \quad (19)$$

Similar to the optimization of $\tilde{\mathcal{D}}_i$, we solve this problem in the Fourier domain, and the optimal solution is

$$\overline{B_1}^{(s)} = \left(2\overline{\tilde{D}_i}^{(s)T}\overline{\tilde{D}_i}^{(s)}+\mu_1 I\right)^{-1} \times\left(\overline{\tilde{D}_i}^{(s)T}\overline{H_1}+\overline{\tilde{D}_i}^{(s)T}\overline{\tilde{X}_i^{(s)}}+\mu_1\left(\overline{A_i^i}^{(s)}+\frac{\Lambda_1}{2\mu_1}\right)\right). \quad (20)$$

Then the optimal $\mathcal{B}_1$ is computed using $\mathcal{B}_1 = ifft(\overline{B_1}, [\,], 3)$.

$\mathcal{A}_i^i$ Subproblem: When optimizing $\mathcal{A}_i^i$, the model is written as

$$\arg\min_{\mathcal{A}_i^i} \mu_1\left\|\mathcal{A}_i^i-\mathcal{B}_1+\frac{\Lambda_1}{2\mu_1}\right\|_F^2 + \mu\left\|\mathcal{A}_i^i\right\|_1 \quad (21)$$

which is solved by the soft threshold algorithm directly

$$\mathcal{A}_i^i = \text{soft}\left(\mathcal{B}_1-\frac{\Lambda_1}{2\mu_1},\frac{\mu}{2\mu_1}\right). \quad (22)$$

Updating Λ_1: The Lagrange multiplier updating formula is as follows:

$$\Lambda_1 = \Lambda_1 + \mu_1\left(\mathcal{A}_i^i-\mathcal{B}_1\right). \quad (23)$$

b) Optimization of $\mathcal{A}_i^j$: Since optimizing $\mathcal{A}_i^j$ with $j\neq i$ needs to fix $\mathcal{A}_i^l$ with $l = 1,\ldots,C, l\neq j$, (15) is rewritten as

$$\arg\min_{\mathcal{A}_i^j} \left\|\tilde{\mathcal{X}}_i-\tilde{\mathcal{D}}_j*\mathcal{A}_i^j-\sum_{l=1,l\neq j}^{C}\tilde{\mathcal{D}}_l*\mathcal{A}_i^l\right\|_F^2 + \left\|\tilde{\mathcal{D}}_j*\mathcal{A}_i^j\right\|_F^2 + \sum_{l=1,l\neq i,j}^{C}\left\|\tilde{\mathcal{D}}_l*\mathcal{A}_i^l\right\|_F^2 + \mu\left\|\mathcal{A}_i^j\right\|_1 + \mu\sum_{l=1,l\neq j}^{C}\left\|\mathcal{A}_i^l\right\|_1. \quad (24)$$

Using $\mathcal{H}_2 = \tilde{\mathcal{X}}_i - \sum_{l=1,l\neq j}^{C}\tilde{\mathcal{D}}_l*\mathcal{A}_i^l$ and removing the irrelevant terms, (24) is simplified as

$$\arg\min_{\mathcal{A}_i^j} \left\|\mathcal{H}_2-\tilde{\mathcal{D}}_j*\mathcal{A}_i^j\right\|_F^2 + \left\|\tilde{\mathcal{D}}_j*\mathcal{A}_i^j\right\|_F^2 + \mu\left\|\mathcal{A}_i^j\right\|_1. \quad (25)$$

This model is also optimized by the splitting strategy to

$$\arg\min_{\mathcal{A}_i^j} \left\|\mathcal{H}_2-\tilde{\mathcal{D}}_j*\mathcal{A}_i^j\right\|_F^2 + \left\|\tilde{\mathcal{D}}_j*\mathcal{A}_i^j\right\|_F^2 + \mu\left\|\mathcal{A}_i^j\right\|_1 \quad \text{s.t. } \mathcal{B}_2 = \mathcal{A}_i^j. \quad (26)$$

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$\mathcal{B}_2$ Subproblem: The optimization model of $\mathcal{B}_2$ is

$$\arg\min_{\mathcal{B}_2} \left\|\mathcal{H}_2 - \tilde{\mathcal{D}}_j * \mathcal{B}_2\right\|_F^2 + \left\|\tilde{\mathcal{D}}_j * \mathcal{B}_2\right\|_F^2 + \mu_2\left\|\mathcal{A}_i^j - \mathcal{B}_2 + \frac{\Lambda_2}{2\mu_2}\right\|_F^2. \quad (27)$$

The optimal solution of $\mathcal{B}_2$ is

$$\overline{B_2}^{(s)} = \left(2\overline{\tilde{D}_j}^{(s)^T}\overline{\tilde{D}_j}^{(s)} + \mu_2 I\right)^{-1} \times \left(\overline{\tilde{D}_j}^{(s)^T}\overline{H_2} + \mu_2\left(\overline{A_i^j}^{(s)} + \frac{\Lambda_2}{2\mu_2}\right)\right). \quad (28)$$

Then the optimal $\mathcal{B}_2$ is computed using $\mathcal{B}_2 = ifft(\overline{B_2}, [\,], 3)$.

$\mathcal{A}_i^j$ Subproblem: The optimization model of $\mathcal{A}_i^j$ is

$$\arg\min_{\mathcal{A}_i^j} \mu_2\left\|\mathcal{A}_i^j - \mathcal{B}_2 + \frac{\Lambda_2}{2\mu_2}\right\|_F^2 + \mu\left\|\mathcal{A}_i^j\right\|_1 \quad (29)$$

which can also be solved by the soft threshold algorithm directly

$$\mathcal{A}_i^j = \text{soft}\left(\mathcal{B}_2 - \frac{\Lambda_2}{2\mu_2}, \frac{\mu}{2\mu_2}\right). \quad (30)$$

Updating Λ_2: The Lagrange multiplier updating formula is as follows:

$$\Lambda_2 = \Lambda_2 + \mu_2\left(\mathcal{A}_i^j - \mathcal{B}_2\right). \quad (31)$$

The procedure of GJTD learning is summarized in Algorithm 2.

Algorithm 2 Procedure of GJTD Learning

Input: $\tilde{\mathcal{X}}$, μ, μ_1, μ_2
Initial: $\Lambda_1 = \Lambda_2 = 0$, $\mathcal{A}$ and $\tilde{\mathcal{D}}$ is initialized by k-SVD, $\mathcal{A}_i^j(j \neq i) = \mathbf{0}$, $iter = 1$
while *not converged and* $iter < MaxIter$ **do**
  for $i = 1{:}C$ **do**
    Update $\overline{\tilde{D}_i}$ by (13);
    Compute $\tilde{\mathcal{D}}_i = ifft(\overline{\tilde{D}_i}, [\,], 3)$.
  end
  for $i = 1{:}C$ **do**
    Update $\overline{B_1}$ by (20);
    Compute $\mathcal{B}_1 = ifft(\overline{B_1}, [\,], 3)$;
    Update $\mathcal{A}_i^i$ by (22);
    Update Λ_1 by (23);
    for $j = 1{:}C$, $j \neq i$ **do**
      Update $\overline{B_2}$ by (28);
      Compute $\mathcal{B}_2 = ifft(\overline{B_2}, [\,], 3)$;
      Update $\mathcal{A}_i^j$ by (30);
      Update Λ_2 by (31).
    end
  end
  $iter \leftarrow iter + 1$.
end
Output: $\tilde{\mathcal{D}}$.

C. Reconstruction Model

Given am LR-HSI, our goal is to get a high-quality HR-HSI using the mapping relationship $\mathcal{D}^l$ and $\mathcal{D}^h$ learned by GJTD. First, we crop the LR-HSI into several overlapping 3-D patches. Second, each patch is assigned into a group. Recall that we have got the cluster center of each group available, and hence we assign each patch into the best fit group by comparing it with all the cluster centers. After that, we can get the tensor form of the LR-his, i.e., $\mathcal{X}^l = [\mathcal{X}_1^l, \ldots, \mathcal{X}_C^l]$.

Two considerations are considered when designing the reconstruction model. For the first one, the desired HR-HSI $\mathcal{X}^h$ can be obtained using the tensor sparse coefficient $\mathcal{A}^*$ and HR dictionary $\mathcal{D}^h$ according to the training phase of GJTD. Nevertheless, in the reconstruction phase, there is no way to get an accurate tensor sparse coefficient $\mathcal{A}^*$ since $\mathcal{X}^h$ is unknown. This is an endless loop in solving the desired HR-HSI $\mathcal{X}^h$ and the tensor sparse coefficient $\mathcal{A}^*$. An alternative way is to estimate the tensor sparse coefficient $\mathcal{A}^*$ using an initial tensor sparse coefficient $\mathcal{A}'$, which is obtained using the LR-HSI $\mathcal{X}^l$ over LR dictionary $\mathcal{D}^l$ [52]. This procedure is shown in the bottom left of Fig. 1.

For the second one, we apply the low-rank prior to regularize $\mathcal{X}^h$ to further capture the spatial–spectral correlation. Many tensor low-rank priors have been proposed recently, such as tensor nuclear norm [53], tensor multirank [5], and weighted tensor nuclear norm [46]. In GJTD-LR, we apply the tensor multirank, which is proposed in the literature [5] and combines the advantages of the tensor nuclear norm and the log-sum of singular values [54]. The log-sum of singular values can shrink the larger singular values of a matrix less to maintain the major useful information and shrink the smaller singular values more to remove the noise. Tensor multirank has similar effect for tensor data. This procedure is shown in the bottom right of Fig. 1.

Based on the two considerations, we first get the initial tensor sparse coefficient $\mathcal{A}'$ using

$$\arg\min_{\mathcal{A}'} \sum_{i=1}^{C}\left(\left\|\mathcal{X}_i^l - \mathcal{D}^l * \mathcal{A}_i'\right\|_F^2 + \left\|\mathcal{X}_i^l - \mathcal{D}_i^l * \mathcal{A}_i'^i\right\|_F^2 + \sum_{j=1, j\neq i}^{C}\left\|\mathcal{D}_j^l * \mathcal{A}_i'^j\right\|_F^2\right) + \mu\left\|\mathcal{A}'\right\|_1. \quad (32)$$

Second, we design the reconstruction model as

$$\arg\min_{\mathcal{X}^h, \mathcal{A}^*}\left(\sum_{i=1}^{C}\left(\left\|\mathcal{X}_i^h - \mathcal{D}^h * \mathcal{A}_i^*\right\|_F^2 + \left\|\mathcal{X}_i^h - \mathcal{D}_i^h * \mathcal{A}^{*i}_i\right\|_F^2 + \sum_{j=1, j\neq i}^{C}\left\|\mathcal{D}_j^h * \mathcal{A}^{*j}_i\right\|_F^2\right)\right) + \tilde{\mu}\left\|\mathcal{A}^*\right\|_1 + \omega\left\|\mathcal{X}^h\right\|_* + \left\|\mathcal{A}^* - \mathcal{A}'\right\|_F^2 \quad (33)$$

where $\tilde{\mu}$ and ω are the regularization parameters of the sparsity term and the low-rank term, respectively. Equation (33) contains four terms. The first and second terms represent that $\mathcal{A}^*$ is an appropriate tensor sparse coefficient for the desired

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HR-HSI $\mathcal{X}^h$ over HR dictionary D^h. The third term represents the tensor multirank for the desired HR-HSI $\mathcal{X}^h$, and the fourth term represents that the initial tensor sparse coefficient $\mathcal{A}'$ is an approximation to the tensor sparse coefficient $\mathcal{A}^*$.

D. Reconstruction Model Optimization

Since the optimization of $\mathcal{A}'$ in (32) and $\mathcal{A}^*$ in (33) are similar to Section III-B2, we mainly introduce the optimization of $\mathcal{X}^h$ in (33), which is separately solved for each cluster, that is,

$$\begin{aligned}\arg\min_{\mathcal{X}_i^h} &\left\|\mathcal{X}_i^h - \mathcal{D}^h * \mathcal{A}_i^*\right\|_F^2 + \left\|\mathcal{X}_i^h - \mathcal{D}_i^h * \mathcal{A}^{*i}_i\right\|_F^2 + \omega\left\|\mathcal{X}_i^h\right\|_* \\ &= \left\|\mathcal{X}_i^h - \frac{\mathcal{D}^h * \mathcal{A}_i^* + \mathcal{D}_i^h * \mathcal{A}^{*i}_i}{2}\right\|_F^2 + \omega\left\|\mathcal{X}_i^h\right\|_*. \quad (34)\end{aligned}$$

To simplify the equation, let $\mathcal{M} = ((\mathcal{D}^h * \mathcal{A}_i^* + \mathcal{D}_i^h * \mathcal{A}^{*i}_i)/2)$, then

$$\arg\min_{\mathcal{X}_i^h} \left\|\mathcal{X}_i^h - \mathcal{M}\right\|_F^2 + \omega\left\|\mathcal{X}_i^h\right\|_*. \quad (35)$$

Based on the definition of $\|\mathcal{X}_i^h\|_*$, (35) is equal to

$$\arg\min_{\overline{X_i^h}^{(s)}} \left\|\overline{X_i^h}^{(s)} - \overline{M}^{(s)}\right\|_F^2 + \lambda LS\left(\overline{X_i^h}^{(s)}\right) \quad (36)$$

where $\overline{X_i^h} = \text{fft}(\mathcal{X}_i^h, [\ ], 3))$, $\overline{M} = \text{fft}(\mathcal{M}, [\ ], 3)$ and $LS(\overline{X_i^h}^{(s)}) = \sum_{j=1}^{\min\{p_1,p_2\}}(\sigma_j(\overline{X_i^h}^{(s)}) + \varepsilon)$. $\sigma_j(\cdot)$ and ε denote the jth singular value and a very small positive number, respectively.

According to [54], $\overline{X_i^h}^{(s)}$ has the closed-form solution $\overline{X_i^h}^{(s)} = U\widetilde{\Sigma}V$. U and V are obtained using SVD on $\overline{M}^{(s)}$, i.e., $\overline{M}^{(s)} = U\Sigma V$. $\widetilde{\Sigma}$ and Σ are two diagonal matrices. The jth diagonal element $\tilde{d}_j$ of $\widetilde{\Sigma}$ is obtained by $\tilde{d}_j = D_{\alpha,\varepsilon}(d_j)$, where d_j is the jth diagonal element of Σ

$$D_{\alpha,\varepsilon}(d_j) = \begin{cases} 0, & c_2 \le 0 \\ \dfrac{c_1 + \sqrt{c_2}}{2}, & c_2 > 0 \end{cases} \quad (37)$$

where $c_1 = |d_j| - \varepsilon$ and $c_2 = c_1^2 - 4(\alpha - \varepsilon|d_j|)$. Then the optimal $\mathcal{X}_i^h$ is computed using $\mathcal{X}_i^h = i\text{fft}(\overline{X_i^h}, [\,], 3)$.

The procedure of the reconstruction model optimization is summarized in Algorithm 3.

Algorithm 3 Procedure of the Reconstruction Model Optimization

Input: The acquired LR-HSI, $\tilde{\mathcal{D}}$, ω, $\tilde{\mu}$, μ
Initial: $iter = 1$
Perform *Tensor grouped transform* operator on the acquired LR-HSI to get $\mathcal{X}^l$.
Compute $\mathcal{A}'$ by optimizing (32).
while *not converged and iter* $<$ *MaxIter* **do**
  Update $\mathcal{A}^*$ by optimizing (33).
  for $i = 1{:}C$ **do**
    Update $\overline{X_i^h}$ by (36) and (37);
    Compute $\mathcal{X}_i^h = ifft(\overline{X_i^h}, [\,], 3)$.
  end
  $iter \leftarrow iter + 1$
end
Perform the *inverse tensor grouped transform* operator on $\mathcal{X}^h$ to get the desired HR-HSI.
Output: The desired HR-HSI.

IV. Experimental Results

In this section, we will test the efficiency of GJTD-LR. The code is available at https://github.com/zhihaofan/GJTD-LR.

A. Experimental Datasets and Implementation Details

To evaluate the performance of GJTD-LR, we use two remote sensed and an indoor HSI datasets as the testing images. The first one is the Washington dc Mall dataset (WDC),[1] which is a remote sensed hyperspectral dataset and acquired from the shopping center in Washington, D.C, using the Hyperspectral Digital Imagery Collection Experiment sensor in 1995. This HSI has 210 bands with the visible spectrum and infrared spectrum ranging from 0.4 to 2.4 μm, and the atmospheric opaque areas in 0.9 and 1.4 μm were omitted from the dataset. The final size is $1280 \times 307 \times 191$, and 1280×307 is the dimension of the spatial modes and 191 is the dimension of the spectral mode. The second one is the Pavia University dataset (PaviaU),[2] which is also a remote sensed hyperspectral dataset and taken from Pavia University by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. This HSI consists of 1096×715 pixels, and each pixel has 115 spectral bands. The third one is the CAVE dataset,[3] which is a widely used indoor hyperspectral dataset and captured by a cooled charge coupled device (CCD) camera from a variety of real-world materials and objects. This dataset consists of 32 indoor HSIs, which can be divided into 5 sections. Each HSI has 512×512 pixels with 31 spectral bands. The spectral bands are acquired by sensors from 400 to 700 nm.

For two remote sensed HSIs, the WDC and PaviaU datasets, we select two subimages containing more image information, with the size of $240 \times 240 \times 191$ and $240 \times 240 \times 115$ as HR-HSIs, respectively. For the indoor HSIs, the CAVE dataset, we first downsample these images by a factor of $\times 2$, which can get the images with the size of $256 \times 256 \times 31$, and then select the center region with the size of $240 \times 240 \times 31$ as HR-HSI. To get the LR-HSI, each HR-HSI is blurred by applying a Gaussian blur of size 5×5 (the mean and standard deviation are 0 and 2, respectively) and is downsampled to four kinds of LR-HSIs of sizes 120×120, 60×60, 40×40, and 30×30 for scaling factors $\times 2$, $\times 4$, $\times 6$, and $\times 8$, respectively. Each LR-HSI is interpolated into the same size of its corresponding HR-HSI. We crop each LR-HSI into many overlapping 3-D image patches of size 5×5 (the

[1] https://engineering.purdue.edu/ biehl/MultiSpec/

[2] http://www.ehu.eus/ccwintco/index.php/

[3] https://www1.cs.columbia.edu/CAVE/databases/multispectral/

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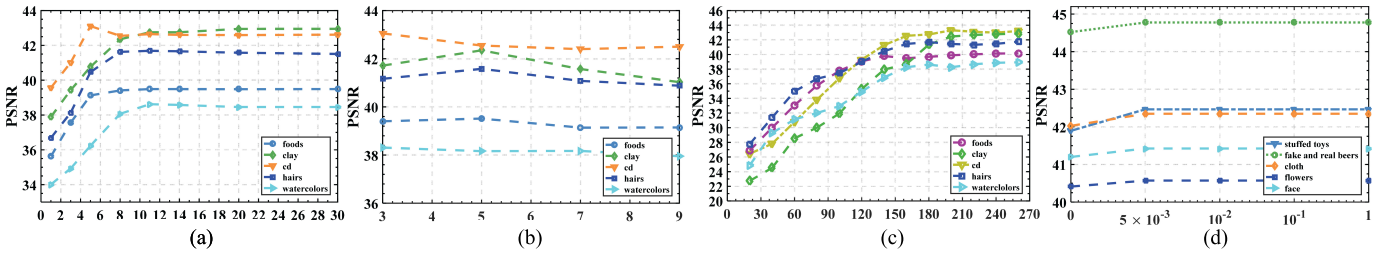


Fig. 3. PSNR curves of (a) number of groups C, (b) size of each patch p, (c) number of atoms k_i, and (d) weight ω.

size of each patch is discussed in Section IV-B) with 3 pixels overlapping. The corresponding HR-HSI is also cropped into many overlapping 3-D image patches using the same way.

Since the CAVE dataset contains 5 different sections, in the training phase, we select 9 HSIs of the CAVE dataset (at least one in each section) as the training set to train the GJTD model. Each HSI is cropped to 1500 image patches. We discard the smooth patches from them and preserve the meaningful patches with a certain amount of edge structures using the intensity variance selection criterion. After that, a total amount of 7131, 6643, 5970, and 5560 image patch pairs are obtained for training with scaling factors $\times 2$, $\times 4$, $\times 6$, and $\times 8$, respectively. In the following experiments, we apply the GJTD trained by 9 HSIs of the CAVE dataset to reconstruct the remaining 23 HSIs of the CAVE dataset. For testing the generality of the proposed GJTD-LR, we also apply the same GJTD to reconstruct the WDC and PaviaU datasets. However, the GJTD contains 31 spectral bands, which is not directly suitable for the WDC and PaviaU datasets because the two datasets contain 191 and 115 spectral bands, respectively. Hence, we divide the whole spectral bands of the WDC and PaviaU datasets into several overlapping groups, respectively, and each group contains 31 spectral bands. For the WDC dataset, we can get 7 overlapping groups, and for the PaviaU dataset, we can get 4 overlapping groups. Each overlapping group is reconstructed independently.

Four widely used quantitative metrics including peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), universal image quality index (UIQI) [55], and relative dimensionless global error in synthesis (ERGAS) [56] are used to evaluate the quality of the reconstructed HSI.

B. Parameters' Discussion

To evaluate the effectiveness and sensitivity of the main parameters in GJTD-LR, we do some experiments to discuss the selection of them. There are the number of clusters C, the size of image patches p (we set $p = p_1 = p_2$), the number of atoms in each subdictionary k_i, and the weight of the low-rank term ω. Ideally, adjusting the parameters for different HSIs can get the best results for them. However, it will reduce the generality of the proposed method. Therefore, in this section, we use the PSNR values of five HSIs in the CAVE dataset (each of them belongs to a section) for the scaling factor $\times 2$ to discuss the selection and sensitivity of the four parameters. Since C, p, and k_i are used in the training phase and ω is used in the reconstruction phase, the displayed HSIs are different in analyzing them. It is noted that the parameters selected are the available values not always the best values.

First, we discuss the selection of C, which has an important impact on the clustering process. Fig. 3(a) shows the PSNR curves obtained using different C from 1 to 30 on foods, clay, cd, hairs, and watercolors (five training HSIs in the CAVE dataset, and each of them belongs to a section). It is noted that $C = 1$ means that the grouped strategy is not carried out, and all the training patches use the same single dictionary. As can be seen from this figure, the PSNR values grow rapidly as C varies from 1 to 8, and they preform stably when C is larger than 8. Hence, we set $C = 8$ in all the experiments. The experimental results show that the grouped strategy improves the reconstruction performance.

Second, we discuss the selection of p, which determines the richness of the inner structures. A larger p can capture rich inner information of the image patch; however, it is also hard to learn the microstructures [14]. Fig. 3(b) shows the plots of the PSNR curves of the reconstructed foods, clay, cd, hairs, and watercolors with different p from 3 to 9. We can see that $p = 5$ is the best selection.

Third, we discuss the selection of the number of atoms in each subdictionary k_i. For simplicity, we set the same number of atoms for different subdictionaries. It is widely known that when the number of atoms is below a certain threshold, the dictionary cannot fully represent the original image structure, leading to unsatisfactory performance. On the contrary, when it is higher than the certain threshold, the PSNR values are stable because the dictionary contains massive redundant atoms. In addition, it will lead to a long training time. Hence, we need to find an appropriate k_i. Fig. 3(c) shows the PSNR curves obtained using different k_i from 20 to 260. This figure indicates that the PSNR values are stable when k_i is larger than 200. Hence, we set $k_i = 200$ in all the experiments.

Finally, we discuss the weight of low-rank term ω, which is only available in the reconstruction phase. Hence, we apply stuffed toys, fake and real beers, cloth, flowers, and face from the testing set in the CAVE dataset (each of them belongs to a section), to analyze the influence of different ω. Fig. 3(d) shows the PSNR curves obtained using different ω from 0 to 1. As can be seen from this figure, the PSNR curves rise rapidly as ω varies from 0 to 0.005, and they do not have obvious changes as ω grows further. Therefore, we choose $\omega = 0.005$ in the experiment. It is noted that $\omega = 0$ means that the PSNR value is obtained by GJTD-LR without the low-rank term. When $\omega = 0$, the PSNR values of stuffed toys, fake and real beers, cloth, flowers, and face are 41.895, 44.522, 42.027, 40.412, and 41.200, respectively. When $\omega = 0.005$, the PSNR values of the five HSIs are 42.467, 44.776, 42.348, 40.571, and 41.425, respectively. From these comparison results,

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we can know that the PSNR values of GJTD-LR with the low-rank term over without the low-rank term are 0.572, 0.254, 0.321, 0.159, and 0.225 dB on five HSIs, respectively.

C. Experimental Results

To test the performance of our proposed GJTD-LR, we compare it with several traditional and advanced methods for single HSI super-resolution. These methods contain the model-based and CNN-based methods. The first method is LRTV [40], which uses a low-rank and TV regularization terms to synthesize the local and global information. The second method is low rank tensor modeling and total variation (LTTV) [42], which uses a tensor nuclear norm and a tensor folded-concave penalty function to describe the global spatial and spectral correlation. The third method is relaxed low tensor train rank (RLTTR) [41], which used the low tensor train rank to learn the spatial–spectral relationship. The fourth method is SSPSR [28], which proposes a spatial–spectral prior network to capture the correlation of spectral bands. The fifth method is GDRRN [33], which is a grouping deep recursive residual network that directly maps the input LR-HSI to HR-HSI through a specialized deep neural network. The sixth method is super-resolution via sparse representation (SCSR) [13], which is a joint dictionary learning super-resolution method for natural images. The seventh method is BicuBic. For a fair comparison, SSPSR, GDRRN, and SCSR also apply the same training and testing datasets with GJTD-LR. Both SCSR and BicuBic are performed band by band, that is, HR-HSI is reconstructed as a series of independent gray images. The parameters required for the training model are trained using the parameters given in their article. All the experiments are performed on a personal computer (PC) with an i9-9900k central processing unit (CPU), 32-GB RAM, and an RTX 2080Ti graphic processing unit (GPU). GJTD-LR, LRTV, LTTV, RLTTR, SCSR, and BicuBic are implemented by MATLAB, while SSPSR and GDRRN are implemented by Python.

1) Result on WDC Dataset: This and Section IV-C2 will verify the effectiveness of GJTD-LR on two remote sensed datasets. Table I lists four quantitative metrics over WDC of all the methods for scaling factors $\times 2$, $\times 4$, $\times 6$, and $\times 8$, where red and blue indicate the best and the second best performance, respectively. We can see that the PSNR improvements of GJTD-LR over the second best methods are 3.58, 3.41, 3.41, and 0.82 dB for scaling factors $\times 2$, $\times 4$, $\times 6$, and $\times 8$, respectively. Two CNN-based methods show unsatisfactory results. Instead, the model-based methods show more satisfactory performance than two CNN-based methods. There might be two reasons: 1) CNN-based methods cannot learn enough feature information with relatively few samples and 2) CNN networks trained using the CAVE training dataset is not suitable for testing the WDC dataset. The proposed GJTD-LR achieves the best performance because it combines the advantages of the learning-based method and the model-based method. Since the first and second spectral bands of WDC have more noises, which are not easy to observe, we adopt the third spectral band to show the reconstructed results. Figs. 4 and 5 show the comparison results of WDC at the third spectral band for scaling factors $\times 2$ and $\times 6$, respectively. As can be seen from Fig. 4, although all the methods can get the reconstructed results, the result obtained by GJTD-LR contains more spatial details with lower errors. When the scaling factor is $\times 6$, the reconstructed results obtained by all the methods have obvious flaws; however, GJTD-LR still achieves acceptable performance as shown in Fig. 5. Although GJTD is trained by the CAVE training dataset, it still shows satisfactory performance on the WDC dataset. The reason mainly comes from three aspects: 1) the dictionary learning can learn the microstructures [14] from relatively small patches. The diversity of different microstructures is guaranteed by relatively few patches; 2) the tensor dictionary is performed on high-frequency feature patches (created by using the first-order and second-order derivatives [13]), which is easy to learn more useful features; and 3) the low-rank prior can alleviate the problem of insufficient samples. Besides that, we display the reconstructed images of WDC for the scaling factor $\times 8$ by various methods to test the effectiveness of different methods when the scaling factor is large. From Fig. 6, we can see that the details of reconstructed images obtained by all the methods are seriously lost. Reconstructed images obtained by RLTTR, LTTV, SCSR, and BicuBic almost lose the entire details. GJTD-LR, LRTV, SSPSR, and GDRRN also lose massive details. Fig. 7 shows the error images of the reconstructed images obtained by GJTD-LR, LRTV, SSPSR,

TABLE I

QUANTITATIVE COMPARISONS OF EIGHT METHODS ON THE WDC DATASET FOR DIFFERENT SCALING FACTORS

		PSNR↑	SAM↓	UIQI↑	ERGAS↓
×2	GJTD-LR	52.3348	3.9593	0.9168	10.2222
	RLTTR [41]	48.1159	9.1624	0.8007	15.8071
	LTTV [42]	47.9451	9.0377	0.7930	16.4095
	LRTV [40]	48.9218	6.5793	0.8694	18.4196
	SSPSR [28]	48.6057	8.3630	0.7786	15.8857
	GDRRN [33]	45.3480	11.5020	0.7329	21.9165
	SCSR [13]	47.3959	8.7597	0.7426	19.3029
	BicuBic	46.5708	10.5797	0.7219	16.9725
×4	GJTD-LR	48.4089	6.1673	0.7851	16.0634
	RLTTR [41]	44.1900	14.5837	0.6217	24.8397
	LTTV [42]	44.0193	14.3957	0.6120	25.7864
	LRTV [40]	44.9959	10.0348	0.6986	28.9451
	SSPSR [28]	44.6798	9.7535	0.5869	24.9633
	GDRRN [33]	41.4221	15.7929	0.3624	34.4402
	SCSR [13]	43.4700	12.0275	0.3570	30.3332
	BicuBic	41.6449	15.7578	0.4059	29.9568
×6	GJTD-LR	47.2986	6.9910	0.7221	18.2539
	RLTTR [41]	43.0796	16.6091	0.5594	28.2270
	LTTV [42]	42.9089	16.4032	0.5498	29.3027
	LRTV [40]	43.8855	11.2968	0.6226	32.8922
	SSPSR [28]	43.5695	14.4241	0.5131	28.3674
	GDRRN [33]	40.3118	17.1287	0.2227	39.1366
	SCSR [13]	42.3596	10.2758	0.2050	34.4695
	BicuBic	40.5346	12.9836	0.2193	24.9509
×8	GJTD-LR	45.8734	7.9413	0.5892	19.6187
	RLTTR [41]	45.0501	10.3625	0.5666	21.2629
	LTTV [42]	44.9770	10.2739	0.5617	21.6326
	LRTV [40]	44.8965	9.5595	0.5368	23.4445
	SSPSR [28]	44.3017	10.8003	0.4261	22.8865
	GDRRN [33]	42.2047	13.4278	0.2630	28.8568
	SCSR [13]	43.1231	10.0572	0.2208	27.7255
	BicuBic	41.5346	11.9836	0.2193	24.9508

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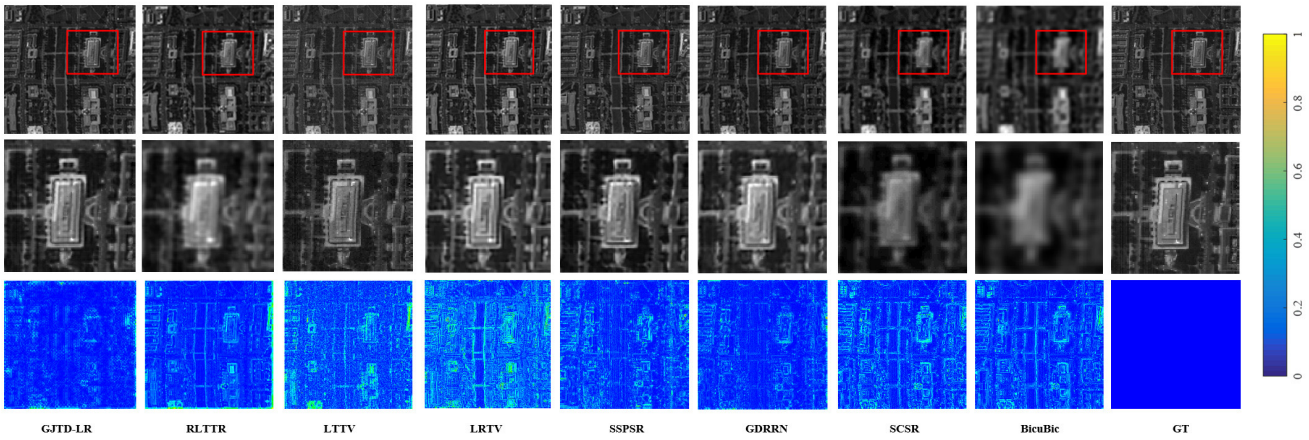


Fig. 4. Reconstructed images, enlarged images, and error images of WDC at the third band for the scaling factor $\times 2$ by various methods. GJTD-LR, RLTTR [41], LTTV [42], LRTV [40], SSPSR [28], GDRRN [33], SCSR [13], BicuBic, and ground truth.

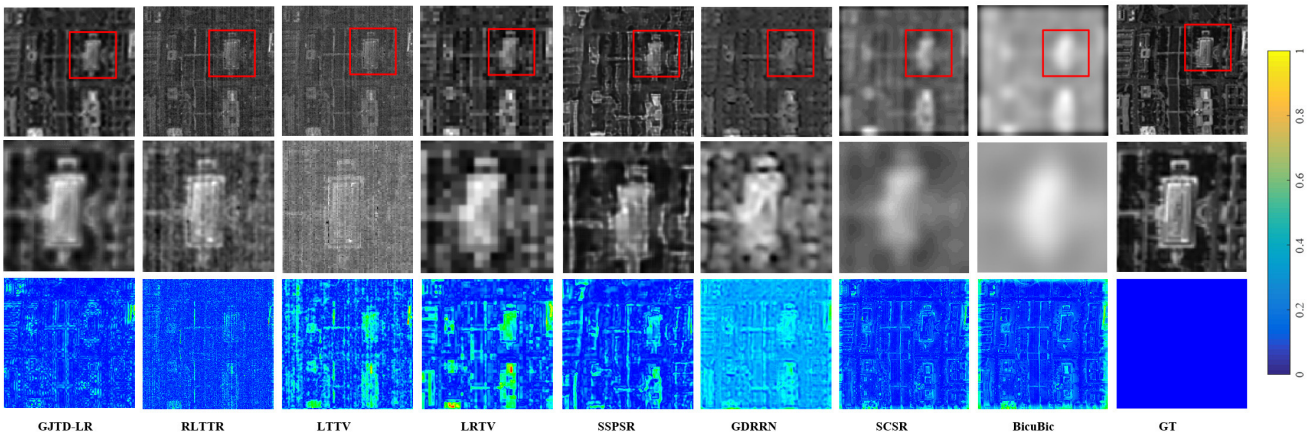


Fig. 5. Reconstructed images, enlarged images, and error images of WDC at the third band for the scaling factor $\times 6$ by various methods. GJTD-LR, RLTTR [41], LTTV [42], LRTV [40], SSPSR [28], GDRRN [33], SCSR [13], BicuBic, and ground truth.

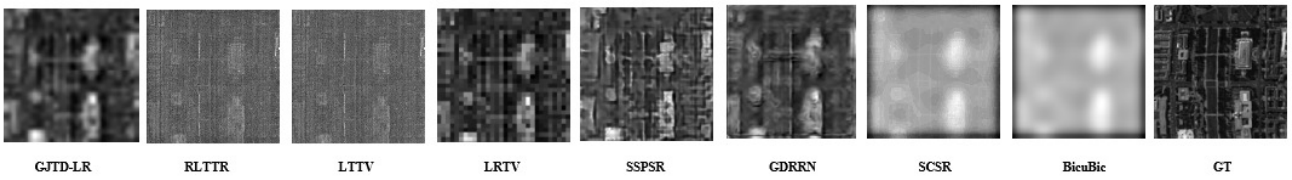


Fig. 6. Reconstructed images of WDC at the third band for the scaling factor $\times 8$ by various methods. GJTD-LR, RLTTR [41], LTTV [42], LRTV [40], SSPSR [28], GDRRN [33], SCSR [13], BicuBic, and ground truth.

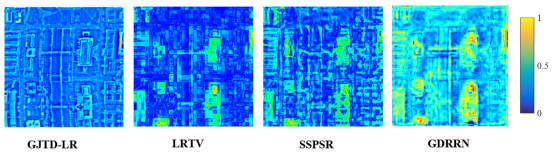


Fig. 7. Error images of the reconstructed images obtained by GJTD-LR, LRTV [40], SSPSR [28], and GDRRN [33] in Fig. 6, respectively.

and GDRRN in Fig. 6, respectively. It can be seen that the error image obtained by GJTD-LR is slightly clearer than that obtained by other methods.

The spectral reflectivity is also an important index for HSI super-resolution. To validate the effectiveness of the reconstructed HSI in the spectral dimension with different methods, we randomly selected a patch with 3×3 from the reconstructed WDC image (scaling factor is $\times 6$) and plotted their average spectral curves as shown in Fig. 8. It is clear that the spectral vector obtained by GJTD-LR is the closest to the ground truth.

2) Result on PaviaU Dataset: Table II lists four quantitative metrics on PaviaU using all the methods for different scaling factors. We can easily observe that GJTD-LR outperforms other methods with respect to all the quantitative metrics for scaling factors $\times 4$, $\times 6$, and $\times 8$. Figs. 9 and 10 show the comparison results of PaviaU at the first band for scaling factors $\times 2$ and $\times 4$, respectively. From the enlarged image

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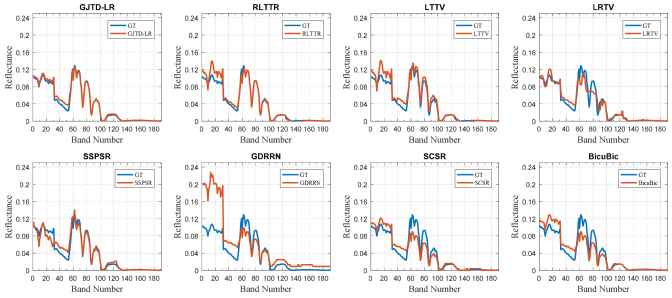


Fig. 8. Average spectral curves of a randomly selected patch with 3 × 3 in the reconstructed WDC for the scaling factor ×6 by different methods. The center of the patch locates at (132, 66).

TABLE II

QUANTITATIVE COMPARISONS OF EIGHT METHODS ON THE PAVIAU DATASET FOR DIFFERENT SCALING FACTORS

		PSNR↑	SAM↓	UIQI↑	ERGAS↓
×2	GJTD-LR	32.9364	3.3797	0.9541	6.6246
	RLTTR [41]	32.5583	5.3653	0.9396	7.2875
	LTTV [42]	32.5601	5.3639	0.9319	7.2681
	LRTV [40]	32.5953	5.3619	0.9379	7.2568
	SSPSR [28]	32.9337	3.3445	0.9550	6.6220
	GDRRN [33]	32.5617	8.3626	0.8396	12.2850
	SCSR [13]	27.9057	7.6021	0.8464	11.9163
	BicuBic	27.9068	8.5956	0.7465	11.9559
×4	GJTD-LR	27.7437	6.0524	0.8468	12.0438
	RLTTR [41]	27.3656	9.4940	0.8256	13.2501
	LTTV [42]	27.0256	9.3476	0.8198	13.2488
	LRTV [40]	26.9802	8.4490	0.6040	17.7663
	SSPSR [28]	27.7409	6.0879	0.8466	12.0440
	GDRRN [33]	24.3667	9.4896	0.4256	20.2486
	SCSR [13]	22.7429	9.7144	0.4602	20.7665
	BicuBic	22.7140	9.7020	0.4011	21.6636
×6	GJTD-LR	25.5905	7.7210	0.7564	16.4424
	RLTTR [41]	25.5060	8.4690	0.7072	18.7861
	LTTV [42]	25.0913	7.7222	0.6564	17.4410
	LRTV [40]	25.5092	8.4711	0.7258	18.7161
	SSPSR [28]	25.2178	7.7494	0.6699	16.4930
	GDRRN [33]	21.5102	13.4718	0.2702	22.7790
	SCSR [13]	22.1815	10.5848	0.2896	23.3343
	BicuBic	22.0606	12.1426	0.2896	23.3336
×8	GJTD-LR	24.2948	8.4278	0.5904	18.0430
	RLTTR [41]	24.0424	8.8003	0.5456	19.4012
	LTTV [42]	23.2945	8.4587	0.5804	18.6417
	LRTV [40]	23.9564	8.8029	0.5346	18.3968
	SSPSR [28]	24.2862	8.4278	0.5893	18.1254
	GDRRN [33]	20.6455	15.8028	0.2346	22.3954
	SCSR [13]	20.9194	13.7910	0.2564	23.7690
	BicuBic	20.3111	13.6551	0.2644	23.7224

of Fig. 9, we can see that the details obtained by GJTD-LR are closest to the ground truth. The result obtained by SSPSR contains too much noises, and the result obtained by RLTTR is fuzzy. In addition, the error image obtained by GJTD-LR is clearer than that obtained by other methods. Fig. 10 gives us the similar conclusion.

3) Result on CAVE Dataset: The experiments on the WDC and PaviaU datasets have confirmed that GJTD-LR has a better performance on the remote sensed HSIs. This section will verify the effectiveness of GJTD-LR on indoor HSIs, CAVE dataset. Since nine HSIs of the CAVE dataset have been used as training images, we use the remaining 23 HSIs

TABLE III

AVERAGE QUANTITATIVE COMPARISONS OF EIGHT METHODS ON THE CAVE DATASET FOR DIFFERENT SCALING FACTORS

		PSNR↑	SAM↓	UIQI↑	ERGAS↓
×2	GJTD-LR	41.9725	1.8088	0.9708	5.1201
	RLTTR [41]	35.6593	4.1831	0.8407	10.2906
	LTTV [42]	35.1720	4.4566	0.8315	10.8605
	LRTV [40]	35.4132	13.1293	0.9296	10.2550
	SSPSR [28]	41.5384	1.8014	0.9647	5.6778
	GDRRN [33]	37.2345	2.9878	0.9444	8.0067
	SCSR [13]	32.2721	3.9394	0.8527	15.3134
	BicuBic	31.2428	3.2532	0.8703	17.0731
×4	GJTD-LR	35.2162	2.8872	0.9368	5.1834
	RLTTR [41]	27.6765	10.2179	0.6618	10.2906
	LTTV [42]	27.8225	10.0282	0.6644	12.4647
	LRTV [40]	29.7706	21.0183	0.8398	9.4466
	SSPSR [28]	34.2232	2.8739	0.9004	6.3883
	GDRRN [33]	30.6039	5.5009	0.8312	8.3517
	SCSR [13]	28.0553	5.4802	0.7497	23.9788
	BicuBic	27.2999	5.0754	0.7529	25.9958
×6	GJTD-LR	32.4864	3.9383	0.8994	4.5973
	RLTTR [41]	22.6813	18.1890	0.5081	15.1217
	LTTV [42]	22.9271	17.6931	0.5140	14.6093
	LRTV [40]	26.5142	26.7107	0.7451	8.9317
	SSPSR [28]	30.8025	3.8302	0.8480	5.8146
	GDRRN [33]	26.7133	8.5476	0.7014	8.5571
	SCSR [13]	25.8901	6.5057	0.6706	28.5686
	BicuBic	25.1855	6.4159	0.6574	30.7859
×8	GJTD-LR	30.9081	4.9088	0.8674	4.0677
	RLTTR [41]	19.6392	24.6767	0.4091	16.0022
	LTTV [42]	19.9218	23.9789	0.4165	15.4052
	LRTV [40]	24.8298	28.9538	0.6674	8.1467
	SSPSR [28]	29.2378	4.6178	0.7887	5.2960
	GDRRN [33]	23.8548	10.6099	0.5841	8.5518
	SCSR [13]	24.8238	7.9660	0.5903	33.9838
	BicuBic	24.2225	8.0447	0.5723	36.2396

as the testing set. Table III reports the average performance over the 23 testing datasets obtained by all the methods for different scaling factors. From the third to sixth columns, we can easily observe that the values of PSNR, UIQI, and ERGAS of GJTD-LR are 0.4 dB higher, 0.06 dB lower, and 0.55 dB lower than the second best method SSPSR, when the scaling factor is ×2. The advantage of GJTD-LR is becoming more and more obvious with the increase in the scaling factor. It should be noted that GJTD-LR did not get the best values of SAM metric, but it still gets the second best values and is very close to the best values.

Fig. 11 shows the PSNR values on the 25 testing images using different methods for different scaling factors. The first 23 labels correspond to the testing images in the CAVE dataset. The 24th and 25th labels correspond to the WDC and PaviaU datasets, respectively. As seen from this figure, GJTD-LR gains the most champions in these testing images.

We select two testing images in the CAVE dataset for visual comparison. Figs. 12 and 13 show the comparison results obtained using various methods on pompoms and cd at the first spectral band for scaling factors ×2 and ×8, respectively. From the error images of Fig. 12 and the enlarged images of Fig. 13, we can see more intuitively that GJTD-LR shows better performance than the compared methods.

We use the reconstructed pompoms for scaling factor ×2 to show the spectral reflectivity of the indoor HSI datasets.

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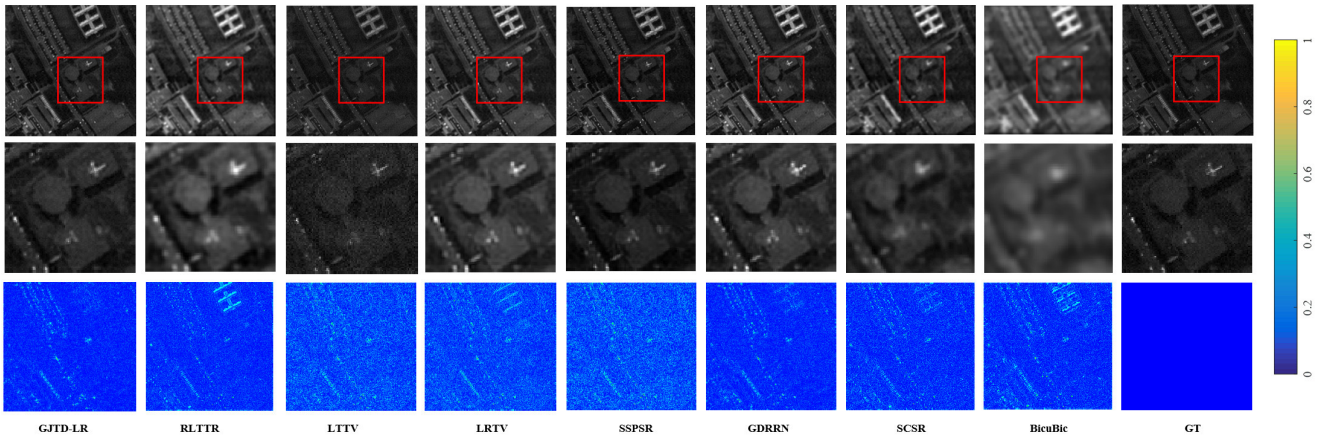


Fig. 9. Reconstructed images, enlarged images, and error images of PaviaU at the first band for the scaling factor $\times 2$ by various methods. GJTD-LR, RLTTR [41], LTTV [42], LRTV [40], SSPSR [28], GDRRN [33], SCSR [13], BicuBic, and ground truth.

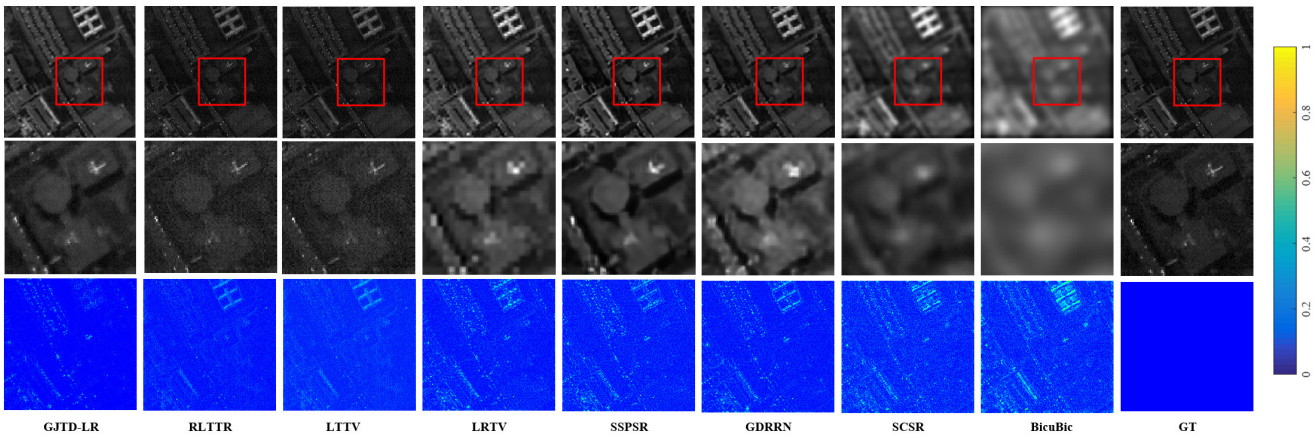


Fig. 10. Reconstructed images, enlarged images, and error images of PaviaU at the first band for the scaling factor $\times 4$ by various methods. GJTD-LR, RLTTR [41], LTTV [42], LRTV [40], SSPSR [28], GDRRN [33], SCSR [13], BicuBic, and ground truth.

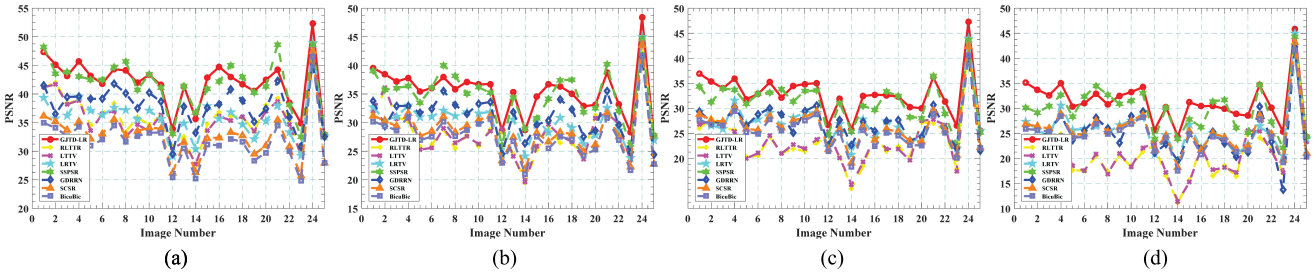


Fig. 11. PSNR values obtained by GJTD-LR over all the testing HSIs (the first 23 HSIs are from the CAVE dataset, and the 24th and 25th are the WDC and PaviaU datasets for scaling factors): (a) $\times 2$, (b) $\times 4$, (c) $\times 6$, and (d) $\times 8$, respectively.

Fig. 14 reveals the spectral cures of a randomly selected patch with 3×3 in reconstructed pompoms using different methods. As displayed in this figure, the proposed GJTD-LR shows a similar performance as SSPSR and outperforms all the other compared methods.

4) Analysis of Model Complexity: The computation time of GJTD-LR is also considered. Since the original codes of SSPSR and GDRRN are based on Python, which have been accelerated by GPU, while the codes of GJTD-LR, RLTTR, LTTV, LRTV, SCSR, and BicuBic are based on MATLAB, we can only make a rough comparison between them. Table IV lists the whole training time on 9 images and average reconstruction time for each image, respectively. "\" represents no this phase. In the training phase, GJTD-LR is faster than SSPSR and SCSR, while it is slower than GDRRN. In the reconstruction phase, GJTD-LR shows a long running time. The principal reason might be that GJTD-LR spends a lot of time in a large quantity of matrix multiplier calculations in the iterative process. How to reduce the computation time is the major problem in future.

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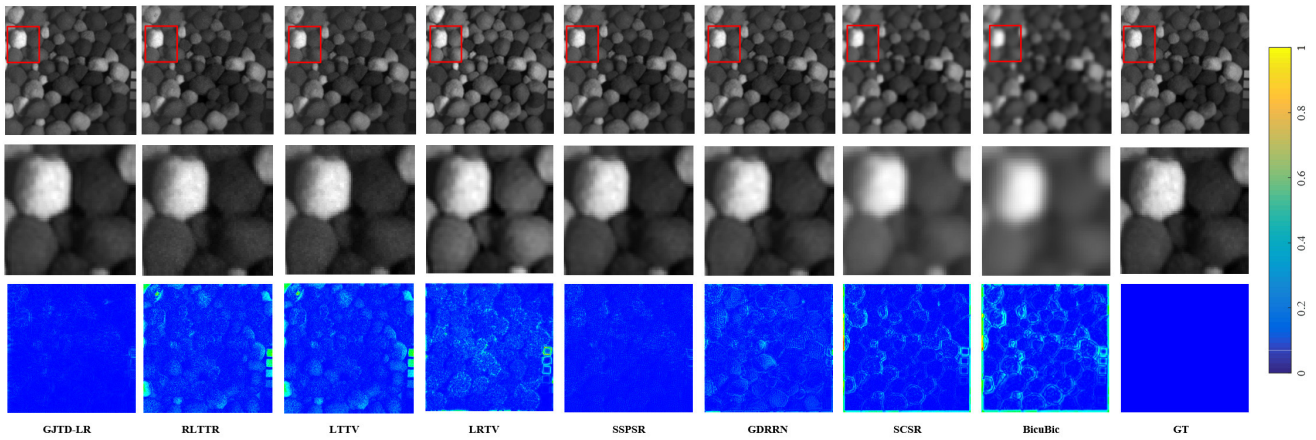


Fig. 12. Reconstructed images, enlarged images, and error images of pompoms at the first band for the scaling factor $\times 2$ by various methods. GJTD-LR, RLTTR [41], LTTV [42], LRTV [40], SSPSR [28], GDRRN [33], SCSR [13], BicuBic, and ground truth.

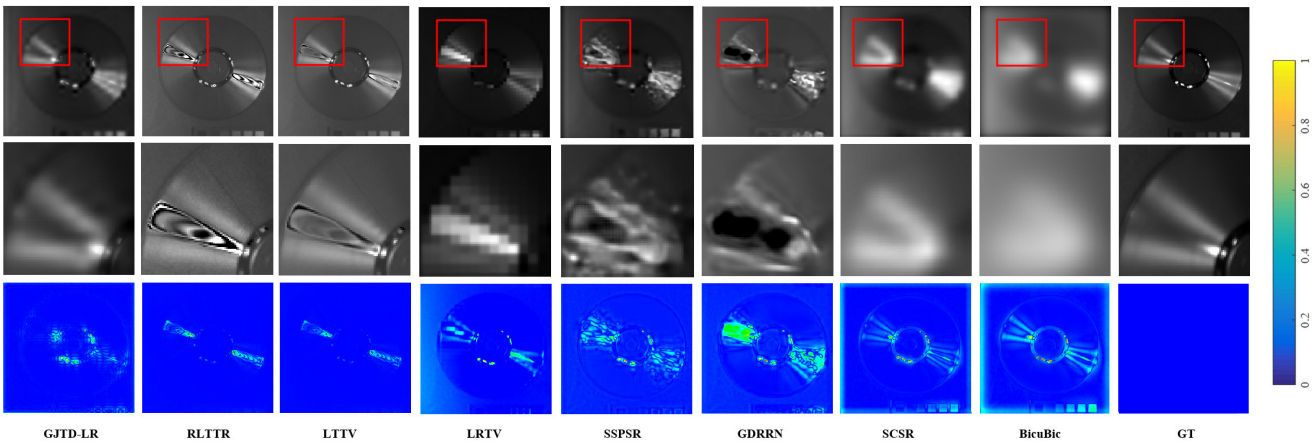


Fig. 13. Reconstructed images, enlarged images, and error images of cd at the first band for the scaling factor $\times 8$ by various methods. GJTD-LR, RLTTR [41], LTTV [42], LRTV [40], SSPSR [28], GDRRN [33], SCSR [13], BicuBic, and ground truth.

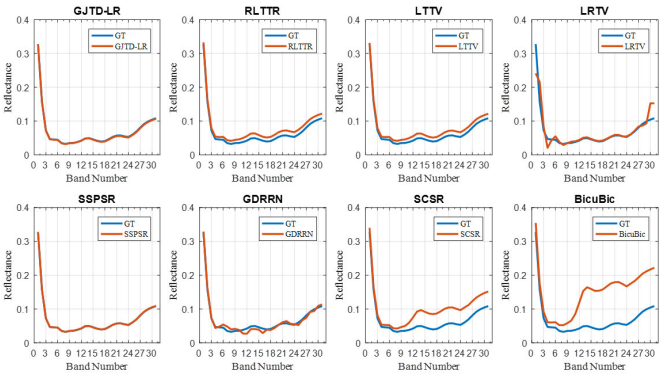


Fig. 14. Average spectral curves of a randomly selected patch with 3×3 in the reconstructed pompoms image (scaling factor $\times 2$) by different methods, respectively. The center of the patch locates at (192, 86).

TABLE IV
COMPUTATION TIME (SECONDS) COMPARISON BETWEEN THE PROPOSED METHOD AND OTHER COMPARED METHODS ON THREE DATASETS

	Training time/s		Average reconstruction time/s	
	$\times 2$	$\times 4$	$\times 2$	$\times 4$
GJTD-LR	8460.10	8140.75	537.39	535.08
RLTTR [41]	\	\	77.83	20.12
LTTV [42]	\	\	58.49	11.47
LRTV [40]	\	\	427.53	387.22
SSPSR [28]	57361.74	48665.06	28.81	25.62
GDRRN [33]	1928.84	1825.57	17.30	15.21
SCSR [13]	12503.75	12467.26	562.08	546.31
BicuBic	\	\	0.083	0.075
	$\times 6$	$\times 8$	$\times 6$	$\times 8$
GJTD-LR	8062.65	7983.75	524.06	519.60
RLTTR [41]	\	\	5.38	5.48
LTTV [42]	\	\	6.26	5.83
LRTV [40]	\	\	352.54	315.41
SSPSR [28]	46657.42	45372.79	23.21	20.88
GDRRN [33]	1506.97	1911.12	12.88	11.06
SCSR [13]	13134.23	12309.22	527.41	502.15
BicuBic	\	\	0.071	0.066

The time complexity of GJTD-LR is analyzed. In the training phase, the main computation burden comes from the K-means clustering method and the optimization of (8). The time complexity of K-means clustering is $O(T_1 C P n)$, where T_1 is the number of iterations in clustering, C is the number of groups, P is the number of dimensions of a latest slice, and n is the number of latest slices. The main computation burden of (8) comes from the matrix multiplier

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calculations of (13), (20), and (28). Due to $C_i > k_i$, $C > P$, and $k_i > P$, the time complexity of (13) is $O(T_2 S m^2 n)$, where T_2 is the number of iterations for optimizing (8). It is noted that both (20) and (28) are mutually exclusive in time complexity, and therefore, we analyze them simultaneously. The time complexity of (20) and (28) is $O(T_2 S m^2 n)$. Hence, the total time complexity of the training phase is $O(T_1 C P n + T_2 S m^2 n)$. In the reconstruction phase, the main computation burden comes from the optimization of $\mathcal{A}'$, $\mathcal{A}^*$, and $\mathcal{X}^h$. The optimization of $\mathcal{A}'$ in (32) and $\mathcal{A}^*$ in (33) is similar to the optimization of $\mathcal{A}$ in Section III-B2. Since (32) has a closed-form solution, its time complexity is $O(S m^2 n)$. $\mathcal{A}^*$ and $\mathcal{X}^h$ in (33) are solved by iteration, so the time complexity of optimizing $\mathcal{A}^*$ is $O(T_3 S m^2 n)$, where T_3 denotes the number of iterations. According to [3], the time complexity of optimizing $\mathcal{X}^h$ in (36) is $O(T_3 S P L n)$, where L denotes the number of singular values. Hence, the total time complexity of the reconstruction phase is $O(T_3 S m^2 n + T_3 S P L n)$.

V. Conclusion

In this article, we proposed a novel single HSI super-resolution method, termed GJTD-LR, to reconstruct an HR-HSI from its corresponding LR-HSI. Aiming at the insufficient training samples; problem of HSIs, GJTD-LR combines the advantages of both the learning-based and model-based methods to improve the performance of the reconstruction result. First, we design a novel GJTD learning model to build the mapping relationship between the HR-HSI and its corresponding LR-HSI with relatively few training samples. Specifically, this model shows three advantages: 1) the dictionary learning can learn the microstructure using relatively small patches, which does not need too much training samples; 2) the grouped dictionary strategy can provide more compact dictionary for each group; and 3) the tensor dictionary can capture the spectral correlation well. Second, we take the low-rank prior into reconstruction model to capture the spatial correlation. An efficient alternating iteration method is developed to optimize GJTD-LR. Massive experiments have demonstrated the advantages of GJTD-LR compared with the traditional and advanced methods.

Future directions are mainly explored from the following four aspects. First, we plan to develop a 4-D GJTD learning model to better capture the spatial and spectral correlations, in which the four dimensions correspond to the two spatial modes, the spectral mode and the grouped mode, respectively. Second, the grouped tensor dictionary strategy can be considered to expand into other tensor factorization models such as tensor train or tensor ring model. Third, it would be worthwhile to try to apply GJTD-LR to other HSI tasks, such as HSI denoising. Finally, how to reduce the time complexity also needs to be considered in future.

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